Immunology and Inflammation

# Beta-Glucan Modulates Monocyte Plasticity and Differentiation Capacity to Mitigate DSS-Induced Colitis

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**eLife Assessment**

This study presents **valuable** and **compelling** evidence that β-glucan-induced trained immunity can protect against intestinal inflammation by reprogramming innate immune cells toward a reparative phenotype. The authors employ a **convincing** combination of functional assays, adoptive transfers, and single-cell transcriptomics to uncover mechanistic insights and demonstrate the therapeutic potential of innate immune memory in IBD. While the work is robust, addressing the underlying epigenetic mechanisms and including additional controls would further reinforce the trained immunity-specific interpretation.

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## Abstract

Trained immunity involves the reprogramming of innate immune cells after an initial exposure, resulting in heightened inflammatory responses to subsequent stimuli and enhanced bactericidal capacity during infection. However, this proinflammatory state could also exacerbate chronic conditions like inflammatory bowel disease (IBD), which is characterized by persistent inflammation and microbial imbalance. It remains unclear how trained immunity influences IBD pathogenesis and whether it can be harnessed therapeutically. In our study, pretreatment with β-glucan reprogrammed bone marrow hematopoietic progenitors and peripheral monocytes, inducing a profound shift in monocyte plasticity and significantly reducing the severity of dextran sulfate sodium (DSS)–induced colitis. Adoptive transfer of bone marrow or peripheral monocytes from β-glucan–trained mice into naive mice conferred robust protection against colitis, demonstrating that this protective effect is transferable. Trained mice also displayed improved clearance of intestinal bacterial infections. Single-cell RNA sequencing revealed an expansion of reparative Cx3cr1$^{+}$ macrophages derived from Ly6C$^{hi}$ monocytes, correlating with accelerated colonic epithelial regeneration. Collectively, these findings reveal how β-glucan–induced trained immunity modulates monocyte differentiation to ameliorate experimental colitis, highlighting the potential of harnessing trained immunity as a therapeutic strategy to recalibrate innate immune responses and restore gut homeostasis in IBD, shedding light for future clinical applications.

## Introduction

Inflammatory bowel disease (IBD), encompassing ulcerative colitis (UC) and Crohn's disease (CD), is a chronic gastrointestinal inflammatory disorder associated with significant morbidity(1, 2). The intestinal mucosa is continuously exposed to a diverse microbial ecosystem, including fungi, bacteria, and viruses(3). In IBD, the delicate balance between the immune system and the microbiota is disrupted, leading to uncontrolled inflammation(4, 5). This dysregulation is further exacerbated by genetic factors, such as loss-of-function mutations in NOD2, which impair antimicrobial peptide production, increase susceptibility to intestinal infections, and decrease IL-10 production by monocytes(6–8).

Trained immunity, a phenomenon where innate immune cells exhibit enhanced responses to secondary challenges, has emerged as a promising therapeutic avenue for various infectious diseases and cancer(9–13). However, its role in chronic inflammatory conditions like IBD remains largely unexplored. While trained immunity can enhance pathogen clearance, it also involves the upregulation of pro-inflammatory cytokines such as TNF, IL-1β, and IL-6(14), therefore the role of trained immunity in IBD remained elusive.

This study aimed to investigate the therapeutic potential of trained immunity in a mouse model of colitis. We hypothesized that inducing trained immunity could enhance microbial control, thereby preserving intestinal integrity and ultimately mitigating disease severity. To test this, we employed β-glucan (BG), a prototypic trained immunity inducer, and assessed its impact on dextran sulfate sodium (DSS)-induced colitis. Our findings demonstrate that BG pretreatment significantly alleviates colitis in mice. Mechanistically, BG induced central immunity in the hematopoietic compartment, leading to the generation of "trained" monocytes in the periphery. Both bone marrow transplantation from trained donors and adoptive transfer of trained monocytes conferred protection against DSS-induced colitis. Furthermore, single-cell RNA-seq revealed that BG-induced trained immunity promotes the expansion of reparative Cx3cr1 intestinal macrophages derived from Ly6C$^{hi}$ monocytes, thereby facilitating epithelial regeneration. In summary, our data suggest that harnessing trained immunity represents a promising therapeutic approach for IBD treatment.

## Results

### β-glucan (BG) pretreatment ameliorates DSS-induced colitis

To investigate the impact of trained immunity on colitis, mice were pretreated with BG, a potent inducer of trained immunity. Successful induction of BG-induced trained immunity (BGTI) was confirmed by the enhanced resistance to *Staphylococcus aureus* infection in BG-treated mice (**Fig S1A** and **S1B**), consistent with previous findings(15). Importantly, BG-pretreatment did not induce spontaneous intestinal inflammation (**Fig S1C** and **S1D**). Following BGTI induction, mice were subjected to DSS-induced colitis. (**Fig 1A**). BG pretreatment significantly ameliorated disease severity, as indicated by the reduced body weight loss (**Fig 1B** and **Fig S1E**), colon shortening (**Fig 1C** and **Fig S1F** and **S1G**), and diminished mucosal inflammation observed via endoscopic imaging (**Fig 1D**). Histological analysis demonstrated lower histopathologic scores (**Fig 1E** and **1F**). Additionally, BG-pretreated mice exhibited enhanced intestinal epithelial barrier integrity, as evidenced by the increased expression of tight junction proteins Zo-1 and Occludin (**Fig 1G** and **1H**), and decreased circulating FITC-dextran levels (**Fig 1I**).

Acknowledging that the long-lasting effects of BGTI(16), we further examined its long-term protective effects against DSS-induced colitis. Despite peripheral myeloid cells and monocytes returning to homeostasis four weeks post-BG treatment (**Fig S2A** and **S2B**), mice remained protected against DSS-induced colitis as demonstrated by body weight change and colon shortening (**Fig 1J** and **1K**), and improved histological outcomes (**Fig S2C** and **S2D**). This BG-induced protection persisted for up to seven weeks (**Fig S2E** and **S2F**). These findings indicate that BG pretreatment effectively ameliorates DSS-induced colitis in mice, emphasizing the therapeutic potential of trained immunity in IBD management.

## BG ameliorates colitis via enhanced myeloid cell activation independent of adaptive immunity

Given that BGTI has been reported to function independently of adaptive immunity(17, 18), we evaluated the effects of BG pretreatment in *Rag1* knockout mice (**Fig S3A-S3D**). Notably, BG pre-treatment remained effective in *Rag1*$^{-/-}$ mice, as evidenced by reduced body weight loss, colon shortening, and improved histological outcomes (**Fig S3E-S3H**). These findings confirm that BGTI-mediated protection is independent of adaptive immune responses, consistent with previous findings demonstrating that adaptive immunity is dispensable for trained immunity.

To gain a comprehensive understanding of the molecular mechanisms underlying BG-mediated protection, we performed RNA sequencing on colonic samples collected at multiple time points following DSS administration. WGCNA identified the MEturquoise module emerged as the core functional module, containing 5,015 genes whose expression exhibited significant co-regulation following BG intervention (**Fig S4A**). And differential gene expression of genes within this module was most significant on day 7 post-DSS treatment when comparing the BG-treated mice to PBS controls (**Fig S4B**). GO analysis revealed significant enrichment in pathways related to chemotaxis and leukocyte migration (**Fig 2A**). KEGG pathway enrichment analysis demonstrated significant activation of cellular processes involved in phagocytic function (**Fig 2B**) Additionally, the enrichment observed in WGCNA was further corroborated by whole-genome analysis of day 7 colitis samples, which also identified significant enrichment of pathways associated with chemotaxis, leukocyte migration and phagocytic function (**Fig S4C** and **S4D**). Multiple analytical approaches concurrently indicated that BG pretreatment not only enhances the recruitment of immune cells but also promotes their functional activation, particularly with respect to phagocytic capabilities.

To further investigate how BG pretreatment reprograms of intestinal myeloid cells, we performed single-cell RNA-seq to characterize intestinal leukocytes populations on day 7 post-DSS treatment. Unbiased clustering analysis identified multiple clusters of intestinal immune cells subsets, including monocytes/macrophages (*Ly6c2*, *Ccr2* and *Adgre1*), dendritic (DC) (*Cst3* and *H2-Aa*), neutrophils (*S100a8/a9* and *Ly6g*), B (*Cd79a Cd79b* and *Cd19*), Pre B (*Myl4* and *Mme*), CD4 T (*Cd3d* and *Cd4*), CD8 T (*Cd3d* and *Cd8a*), natural killer (*Nkg7*), ILC2 (*Gata3* and *Il4*) and ILC3 (*Rorc* and *Il22*) cells (**Fig 2C** and **Fig S4E**). Importantly, the ratio of monocytes/macrophages and neutrophils was elevated in the BG group (**Fig 2D**).

Single-cell RNA-sequencing data further demonstrated enhanced expression of innate pathogen recognition receptors (*Syk*, *Nod1*, *Nod2*, *Tlr2* and *Tlr4*), efferocytosis (*Cd300lf*, *Axl*, *Mertk*, *Itgal* and *Itgb7*), cytokines (*Il1b*, *Il1rn*, *Vegfa* and *Tgfb1*), chemokines (*Ccl2*, *Ccl6*, *Ccl9*, *Cxcl9*) and chemokine

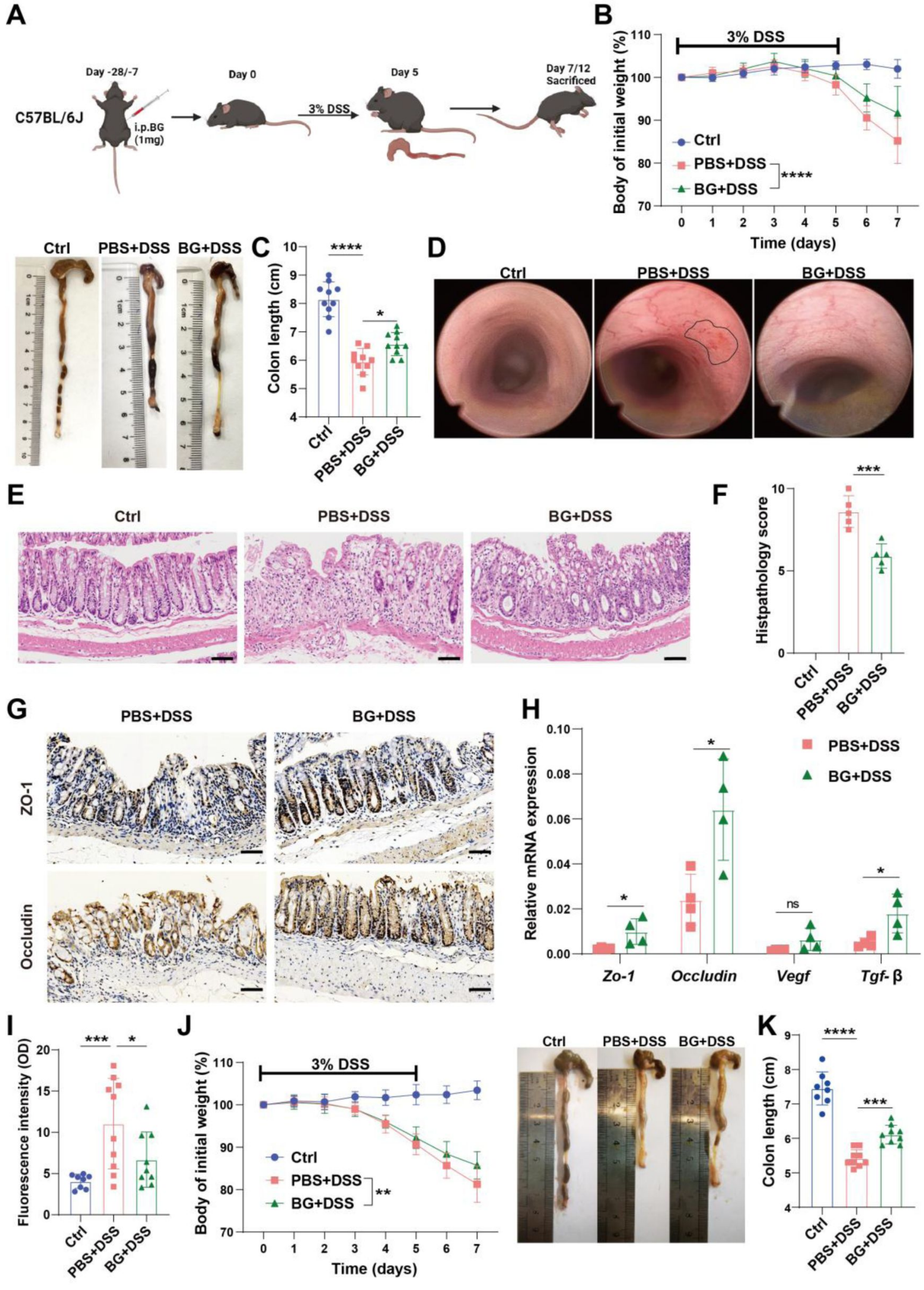
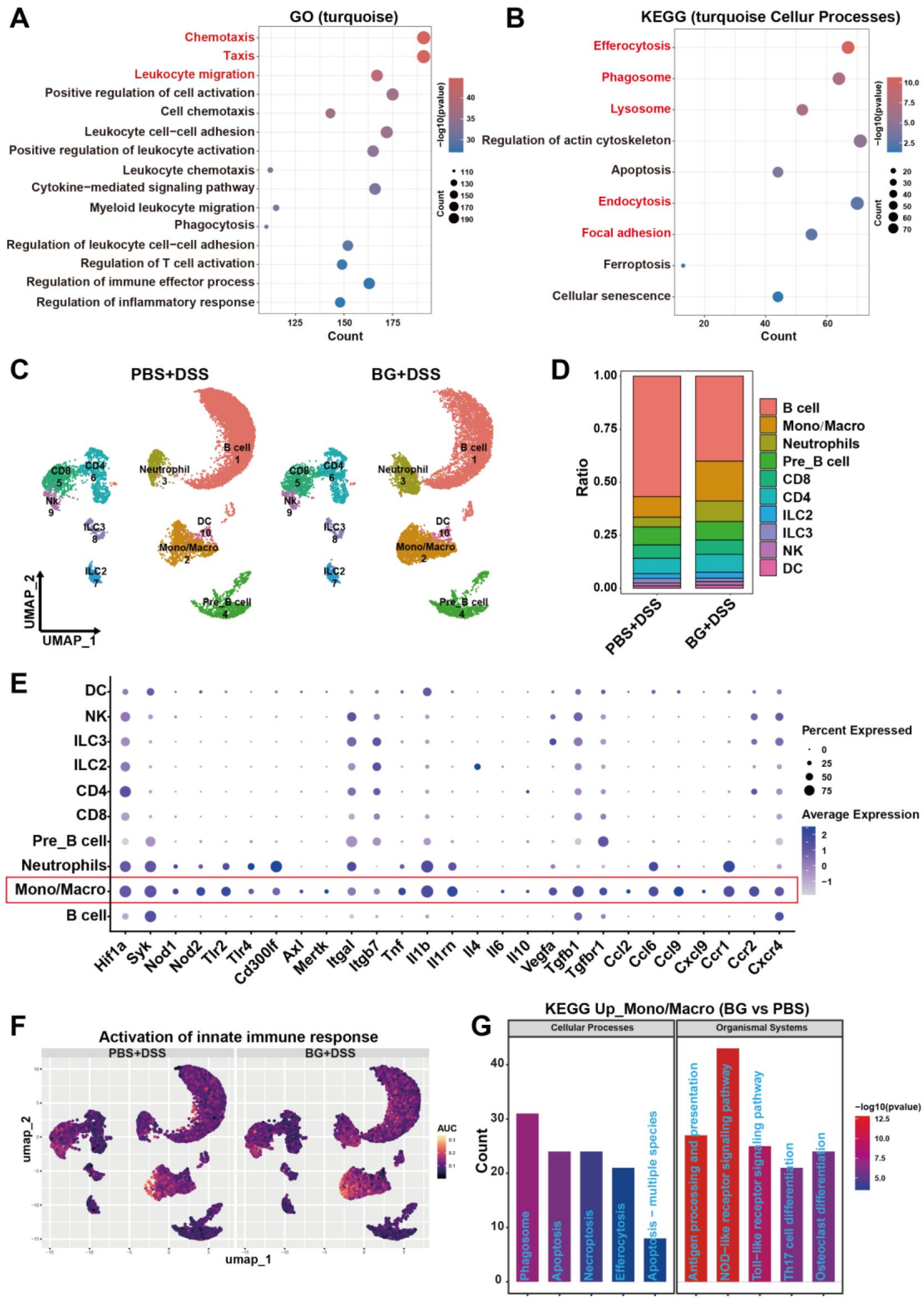



**Fig 1.**

### β-glucan (BG) pretreatment ameliorates DSS-induced colitis.

(A) Schematic representation of BG-induced trained immunity and DSS colitis model, (**B**) Body weight change curve of mice pretreated with BG for one week, followed by colitis induction with 3% DSS ($n$ = 10), (**C**) Colon length changes in mice from (**B**), (**D**) Endoscopic images displaying mucosal damage, (**E and F**) H&E staining and histological scoring. Scale bars: 100 μm, (**G and H**) Expression levels of tight junction and repair, (**I**) FITC-dextran assay assessing intestinal barrier function, (**J**) Body weight change curve of mice pretreated with BG for four weeks, followed by colitis induction with 3% DSS ($n$ = 8-9), (**K**) Colon length changes in mice from (**J**). Data are presented as mean ± SD. Statistical significance: *$p$ < 0.05, **$p$ < 0.01, ***$p$ < 0.001, ****$p$ < 0.0001. ns, not significant.




## Fig 2.

### BG ameliorates colitis by enhancing myeloid cell activation.

RNA sequencing of colon tissue at different time points of colitis. (**A**) GOBP and (**B**) KEGG pathway analyses of genes in the MEturquoise module. Single-cell RNA sequencing analysis of CD45$^+$ cells in the colon on day seven of colitis after one week of BG pretreatment. (**C**) UMAP plot of LP CD45$^+$ cells, (**D**) Cell ratio distribution from scRNA-seq data, (**E**) Dot plots showing representative DEGs between LP CD45$^+$ cells, (**F**) AUC scores for selected pathways, (**G**) KEGG pathway analysis of genes upregulated in the monocyte-macrophage lineage. DEGs, differentially expressed genes. LP, lamina propria. UMAP, uniform manifold approximation, and projection. AUC, area under the curve.

receptors (*Ccr1*, *Ccr2*, *Cxcr4*) in intestinal monocytes/macrophages (**Fig 2E**). AUC and KEGG analysis demonstrated significant enrichment pathways related to activation of innate immune response and phagocytosis in monocytes/macrophages from BG-treated mice (**Fig 2F** and **2G**). These findings, corroborated by WGCNA analysis, strongly suggest that BG pretreatment enhanced the recruitment and activation of myeloid cells, particularly monocytes/macrophages, in the inflamed colon, thereby improving disease outcomes.

## BG trained bone marrow monocytes protected against colitis via the Ccl2-Ccr2 axis

Drawing upon single-cell transcriptome analysis that reveals the proportional expansion and functional reprogramming features of colonic monocyte/macrophage populations, in conjunction with the observation of significantly upregulated peripheral monocytes following BG pretreatment as depicted in **Fig S2B**. A hypothesis was formulated that BG exerts protective effects against colitis via monocytes. To experimentally validate this hypothesis, we evaluated the role of monocytes in *Ccr2* KO mice. Circulating Ly6C$^{hi}$ monocytes were diminished in *Ccr2*$^{-/-}$ mice (**Fig S5A** and **S5B**), corroborating the pivotal role of the Ccl2-Ccr2 axis in monocyte egression from the bone marrow into circulation(19, 20). Notably, BG pretreatment failed to protect *Ccr2*$^{-/-}$ mice from DSS-induced colitis, as evidenced by the lack of significant differences in body weight loss, colon length, and epithelial permeability (**Fig 3A-E**).

In view of the fact that BG induces myelopoiesis through the modulation of hematopoietic stem and progenitors cell compartments in the bone marrow via central trained immunity(21, 22). we assessed whether transplantation of bone marrow cells from BG-pretreated donors could confer resistance to DSS-induced colitis in naïve recipients (**Fig 3F**). At 6 weeks post-transplantation, circulating myeloid cells were predominately derived from donor mice as indicated by the CD45.1 marker on circulating mononuclear cells (**Fig S5C**). Following DSS treatment, mice receiving bone marrow cells from BG-pretreated donors exhibited significantly reduced body weight loss (**Fig 3G**). However, no significant difference in colon length was observed (**Fig 3H** and **Fig S5D**). Meanwhile, the percentage of circulating CD11b$^{+}$ myeloid cells, neutrophils and Ly6C$^{hi}$ monocytes were increased (**Fig 3I and J**, **Fig S5E**).

To further assess the role of monocyte, we adoptively transferred bone marrow Ly6C$^{hi}$ monocytes sorted from control or BG-trained donor mice into *Ccr2*$^{-/-}$ recipient mice, which were then subjected to DSS treatment (**Fig S5F** and **S5G**). Mice received BG-trained monocytes exhibited reduced weight loss and colon shortening (**Fig 3K** and **3L**). These findings suggest that BG-trained Ly6C$^{hi}$ Ccr2$^{+}$ monocytes play a pivotal role in the alleviation of colitis.

## BG-trained monocytes enhance innate immune activation and microbial control

To explore the heterogeneity and functional roles of monocytes and macrophages, we delved deeper into the scRNA-seq data. By examining the expression of key signature genes, including *Ly6c2*, *Ccr2*, *H2-Ab1*, *Runx3*, *Itgax*, *Adgre1*, *Cx3cr1* and *Cd209a*, we identified eight distinct immune cell subsets: three monocyte subsets (Mono1-3) characterized by high *Ly6c2* and *Ccr2* expression, four macrophage subsets (Macro1-4) defined by expression of *H2-Ab1*, *Runx3*, *Itgax*, *Adgre1*, and *Cx3cr1*, and a dendritic cell subset (Cd209a$^{+}$DC) expressing integrin *Itgax* and C-type lectin *CD209a* (**Fig 4A** and **Fig S6A**). Among macrophage subsets, Macro1 highly expressed *Vegfa*, suggesting that may be a mucosal repair-promoting macrophage population. Macro2 was enriched with tissue-resident macrophage signature genes (*Runx3*, *Dtx4*, *Cx3cr1*, *Hes1*). In contrast, Macro3 and Macro4 deviated from the main cluster in cell clustering analysis, showing significant heterogeneity. Macro3 was characterized by high *Ighm* expression, while Macro4 showed elevated *Cd4* expression. Both Macro2 and Macro4 exhibited higher *Il10* expression than other monocyte/macrophage subsets, indicating their inflammation-regulatory functions.

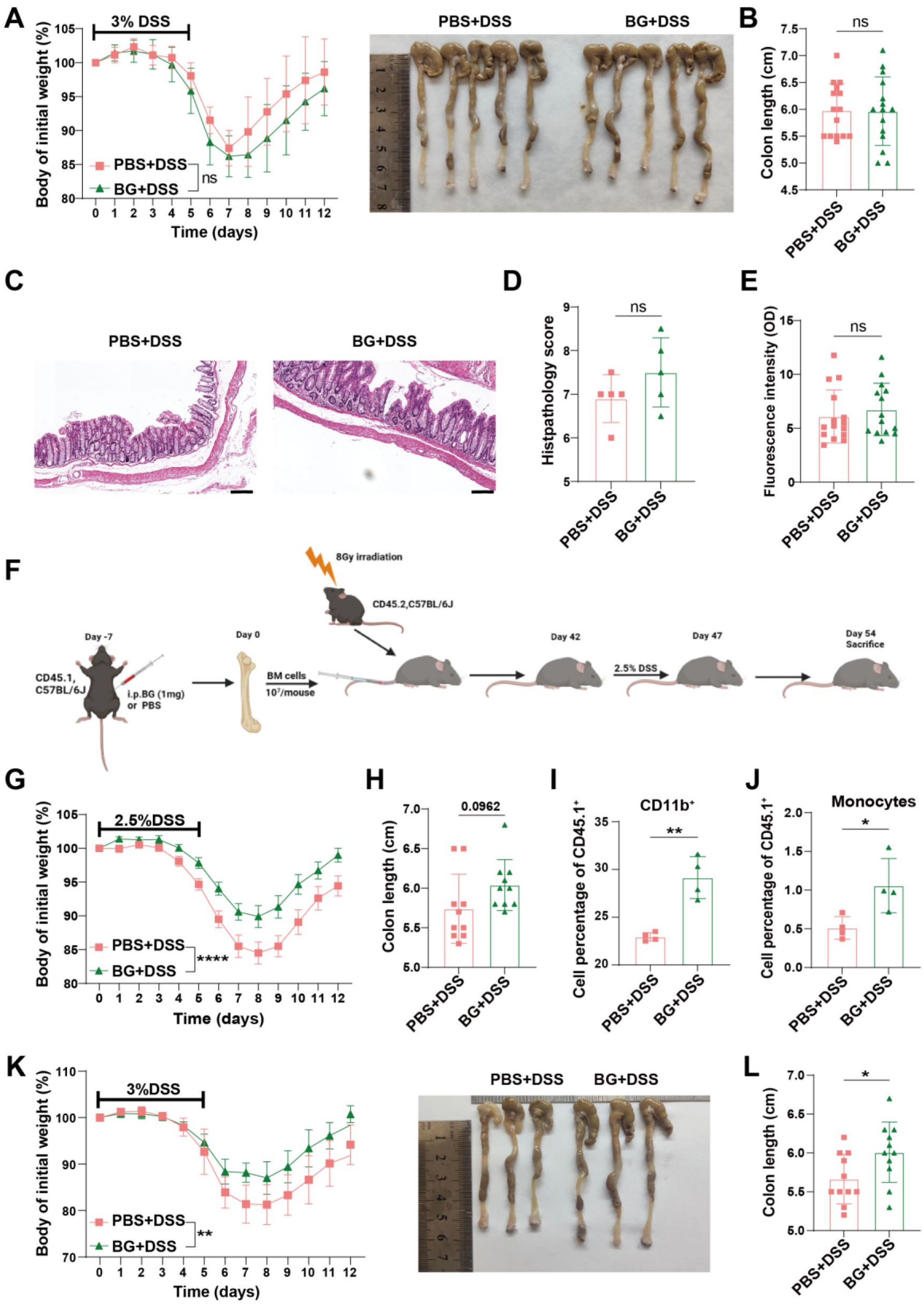



**Fig 3.**

**BG trained bone marrow monocytes protected against colitis via the Ccl2-Ccr2 axis.**

*Ccr2*$^{-/-}$ mice pretreated with BG for 1 week followed by induction of colitis with 3% DSS. (**A**) Changes in body weight ($n$ = 15), (**B**) Colon length changes in *Ccr2*$^{-/-}$ mice from (**A**), (**C and D**) H&E staining and histological scoring, H&E. Scale bars: 100 μm, (**E**) FITC-dextran assay assessing intestinal barrier function, (**F**) Schematic representation of bone marrow transplantation from BG-pretreated CD45.1 mice to CD45.2 mice and colitis model, (**G**) Body weight change curve of colitis mice ($n$ = 10), (**H**) Colon length changes in colitis mice from (**G**), (**I and J**)Percentage of CD11b$^{+}$ (**I**) and Ly6C$^{hi}$ monocytes (**J**) were analyzed by flow cytometry in peripheral blood, (**K and L**) Body weight change curve(**K**) and colon length changes of *Ccr2*$^{-/-}$ mice receiving Ly6C$^{hi}$ monocyte adoptive transfer(**L**) ($n$ = 11). Data are presented as mean ±SD. Statistical significance: *$p$ <0.05, **$p$ <0.01, ****$p$ <0.0001. ns, not significant.

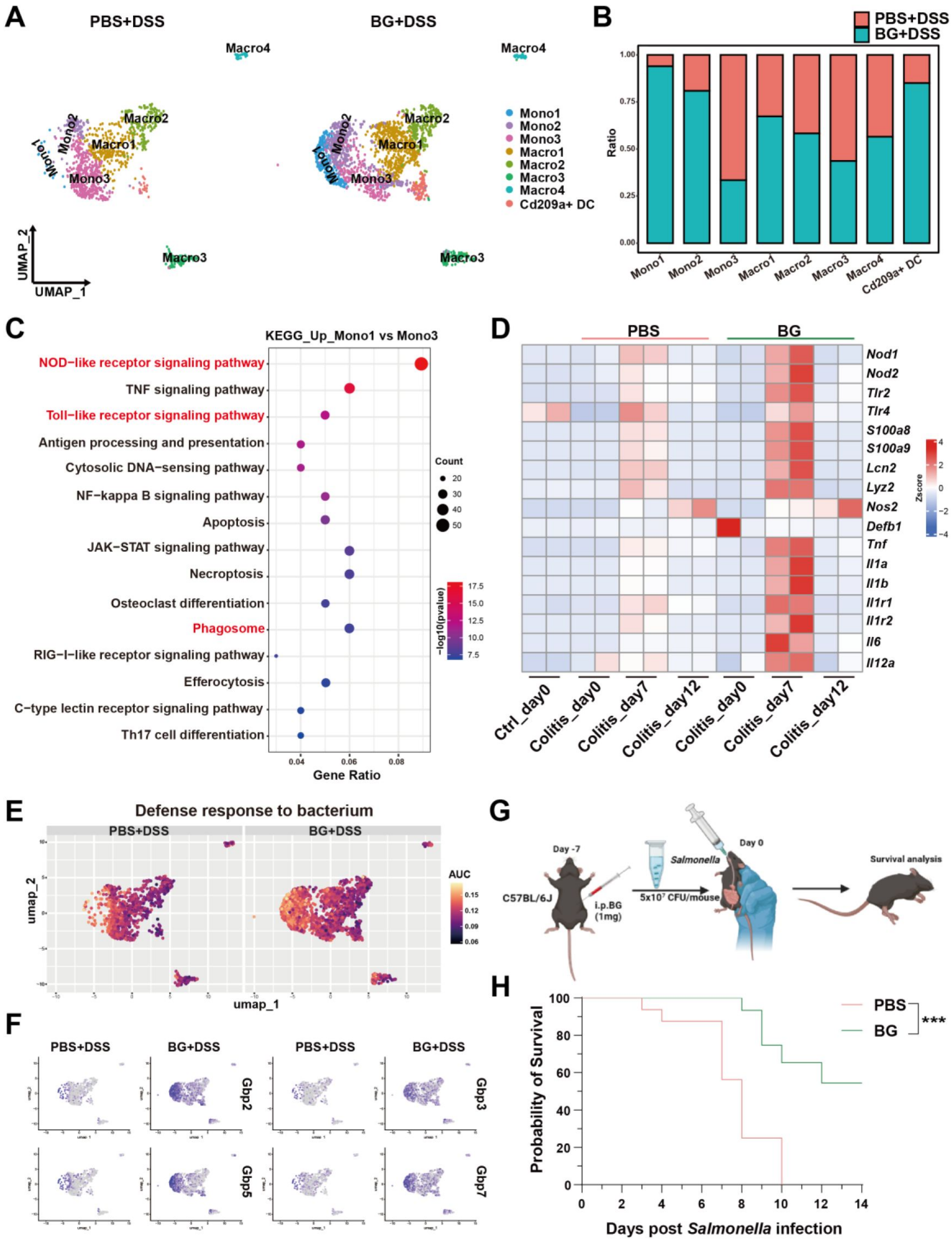



**Fig 4.**

### BG-trained monocytes enhance innate immune activation and microbial control.

(**A**) UMAP and graphical visualization of the monocyte/macrophage lineage, (**B**) The ratio of monocyte/macrophage subsets, (**C**) KEGG pathway enrichment analysis of genes upregulated in monocyte 1 and monocyte 3, (**D**) Gene expression analysis at different time points of colitis, based on colon RNA sequencing, (**E**) AUC scores of selected pathways, (**F**) UMAP plots showing differential gene expression patterns, (**G**) Schematic representation *Salmonella* resistance after one week of BG pretreatment, (**H**) Survival curve of mice infected with *Salmonella* after one week of BG training. Statistical significance: ***$p < 0.001$.

The BG-treated group exhibited a significant increase in the proportions of Mono1, Mono2, Macro1, and CD209a$^{+}$DCs, whereas higher proportions of Mono3 and Macro3 were observed in the PBS group (**Fig 4B**). Given the strong enrichment of Mono1 and the reduction of Mono3 following BG treatment, we sought to further define the transcriptomic characteristics that differentiate Mono1 from Mono3. GO enrichment analysis revealed that Mono1 was significantly enriched in pathways regulating the activation of innate immune response (**Fig S6B**). KEGG pathways analysis further identified enrichment in NOD-like receptor (NLR) signaling, Toll-like receptor (TLR) signaling, and phagocytosis (**Fig 4C**). AUC analysis further confirmed the TLR and NLR signaling pathways were significantly upregulated in Mono1 and Mono2 subsets of the BG-pretreated group (**Fig S6C** and **S6D**). Additionally, antimicrobial humoral response pathway was also upregulated in these two subclusters (**Fig S6E**). Cooperative activation of the pattern recognition receptor (PRR) signaling axis and antimicrobial humoral response significantly promotes the synthesis of antimicrobial peptides(23, 24). BG pretreatment significantly upregulated genes encoding pattern recognition receptors (*Nod1*, *Nod2*, *Tlr2*, *Tlr4*) and antimicrobial effector molecules (*S100a8*, *S100a9*, *Lcn2* and *Defb1*) in the colon on day 7 of colitis (**Fig 4D**). Furthermore, antimicrobial genes *S100a8* and *Defa1* were highly expressed in the BG group as validated by qPCR analysis (**Fig S6F**). These genes upregulation suggest a potential link between trained immunity and enhanced microbial control. Consistently, gene signatures associated with bacterial defense responses were prominently enriched in the BG-treated group (**Fig 4E**).

Previous studies have demonstrated that GBPs as intracellular effectors induced by IFNγ and LPS to promote antibacterial defense(25, 26). We found that BG training significantly enriched the IFN-γ response pathway in Mono1 and Mono2 subsets (**Fig S6G**). Meanwhile, Guanylate-binding proteins (GBP)-related genes, including *Gbp2*, *Gbp3*, *Gbp5*, and *Gbp7*, were also specifically upregulated in Mono1 and Mono2 (**Fig 4F**). GBP-deficient mice exhibit significantly increased susceptibility to *Salmonella typhimurium* infection(27–29). We then performed intestinal *Salmonella typhimurium* infection to assess whether BG pretreatment enhances microbial control in the gut (**Fig 4G**). As expected, BG pretreatment significantly protected mice from lethal *Salmonella Typhimurium* infection, suggesting that BG-induced trained immunity enhances mucosal defenses against gut microbial infections (**Fig 4H**).

Taken together, these results reinforced our bulk RNA-seq findings, indicating that BG pretreatment induces a specific monocyte subcluster with enhanced innate activation and phagocytic capacity, facilitating more efficient control of the breaching microbiome associated with DSS-induced leaky gut.

## BG-mediated reprogramming of myeloid differentiation trajectories balances inflammation and enhances mucosal repair in colitis

While BG-induced trained immunity is characterized by enhanced pro-inflammatory cytokine production and increased bactericidal capacity(14), excessive pro-inflammatory cytokines release may exacerbate colitis progression. To balance heightened cytokine production without inducing overwhelming immunopathology, we hypothesized that BG pretreatment might involve feedback loops that modulate the inflammation while preserving enhanced phagocytic capacity.

To investigate this, we analyzed previously obtained bulk colonic RNA-seq data, focusing on genes associated with anti-inflammatory and regulatory functions. On day 7 of colitis, BG pretreatment significantly upregulated genes encoding immunomodulatory factors (*Il4*, *Il10*, *Il11* and *Il33*), signaling inhibitors (*Socs1*, *Socs3*), and differentiation regulators (*Csf1*, *Csf1r*, *Stat3* and *Stat6*) (**Fig S7A**) The upregulation of *Socs1* and *Socs3* was further corroborated by scRNA-seq data, which showed increased expression of these genes in the Mono1 and Mono2 subclusters (**Fig S7B**). As members of the suppressor of cytokine signaling (SOCS) family, SOCS1 and SOCS3 inhibit excessive

JAK-STAT signaling activation via negative feedback regulation(30–32). Moreover, coordinated activation of the monocyte/macrophage differentiation regulatory network (*Csf1r*, *Nr4a2*, *Irf8*, *Klf4*) suggests that BG may induce the differentiation of regulatory-phenotype macrophage subsets by reprogramming myeloid developmental trajectories, thereby modulating inflammatory responses.

To further investigate whether BG pretreatment reprograms monocyte-to-macrophage differentiation, we analyzed the day7 RNA-seq data revealed significant enrichment of myeloid leukocyte differentiation pathway (**Fig S7C**). Additionally, Single-cell trajectory analysis further revealed that Macro1 and Macro2 differentiated from Mono2 and Mono3, respectively (**Fig 5A**). BG training significantly accelerated the differentiation kinetics of Mono3 into Macro1, which may underlie the reduced frequency of the Mono3 subset in the BG group. Coupled with the gene expression signatures in **Fig S6A**, Macro1 shares similarities with Macro2 in marker gene expression, and the differentiated Macro1 subset could further differentiate into the tissue-resident macrophage Macro2 subset (**Fig 5B**).

Furthermore, we observed that the expression levels of *Ccr2* and *Ly6c2* decreased from Mono1 to Macro2, while *H2-Ab1* and *Cx3cr1* progressively increased, particularly in macrophages (**Fig 5C**). Since Cx3cr1 upregulation is a hallmark of circulating monocytes migrating to the intestinal lamina propria and undergoing macrophage differentiation(33), we used *CX3CR1-GFP* reporter mice to monitor the monocyte-to-macrophage transition in the DSS colitis model. In circulating leukocyte subsets, the total percentage of CD11b$^+$ myeloid cells was higher in BG-pretreated mice compared to controls (**Fig S8A-C**). A similar trend was observed in neutrophils (**Fig S8D** and **S8E**). While the initial monocyte percentage was higher in the BG group, no significant difference was observed the two groups on day 7 post-DSS treatment (**Fig 5D** and **S8F**), suggesting that circulating monocytes in BG-pretreated mice had extravasated into the intestinal tissue. Meanwhile, the percentage of colonic CD11b$^+$ myeloid cells and neutrophils was increased in the BG-pretreated group (**Fig S9A-C**).

We further defined monocyte/macrophage populations from P1 to P6 based on the expression of CD11b, CX3CR1-GFP, Ly6C, and MHCII. P1 to P3 were gated from CD11b$^+$&CX3CR1-GFP$^-$&Ly6G$^-$ cells. While P1 (Ly6C$^+$&MHCII$^-$) and P3 (Ly6C$^-$&MHCII$^+$) were comparable between group, P2 (Ly6C$^+$&MHCII$^+$) was increased in BG-pretreated mice (**Fig 5E**). P4 to P6 are gating from CD11b$^+$&CX3CR1-GFP$^+$ cells and further separate by the expression of Ly6C and MHCII. P4 (Ly6C$^+$&MHCII$^-$) as infiltrating monocytes gaining CX3CR1 expression. P5 (Ly6C$^+$&MHCII$^+$) is regarded as intermediate transitory immature macrophage and P6 (Ly6C$^-$&MHCII$^+$) is considered as mature macrophage as it completely loss expression of Ly6C and highly express MHCII. BG pretreatment led to an increase in P5 level (**Fig 5F**). The increase of P2 and P5 in BG group in line with the decrease of monocytes at day 7 in circulation post DSS treatment, suggesting that BG trained monocytes have higher capacity to infiltrate into the damaged colon and undergo macrophage differentiation.

Previous WGCNA analysis suggested that BG pre-treatment upregulated the focal adhesion pathway, which may enhance tissue repair mechanisms (**Fig 2B**). Consistent with this, we observed increased expression of genes involved in wound healing and tissue repair, such as *Mmp*-related genes, *Spon1*, *Notch1*, *Notch2*, *Tgfb*, *Fgf1*, *Fgf2*, *Col1a1*, *Col1a2, and Fn1* in the BG-treated group on day 7, followed by a rapid decline by day 12. In contrast, the PBS group exhibited a much milder induction of these genes, with several genes remaining at comparable levels between day 7 and day 12 (**Fig 5G**). This led us to hypothesize that the upregulated monocytes and macrophages in the BG-treated group might exhibit enhanced expression of tissue repair-associated genes. As expected, the expression of *Spon1* and *Mmp14*, which promote the mucosal repair was increased in Mono2, Macro1, and Macro2 (**Fig 5H** and **5I**). These findings suggest that BG pretreatment indeed induced monocyte and macrophage subsets with enhanced tissue repair capacity.

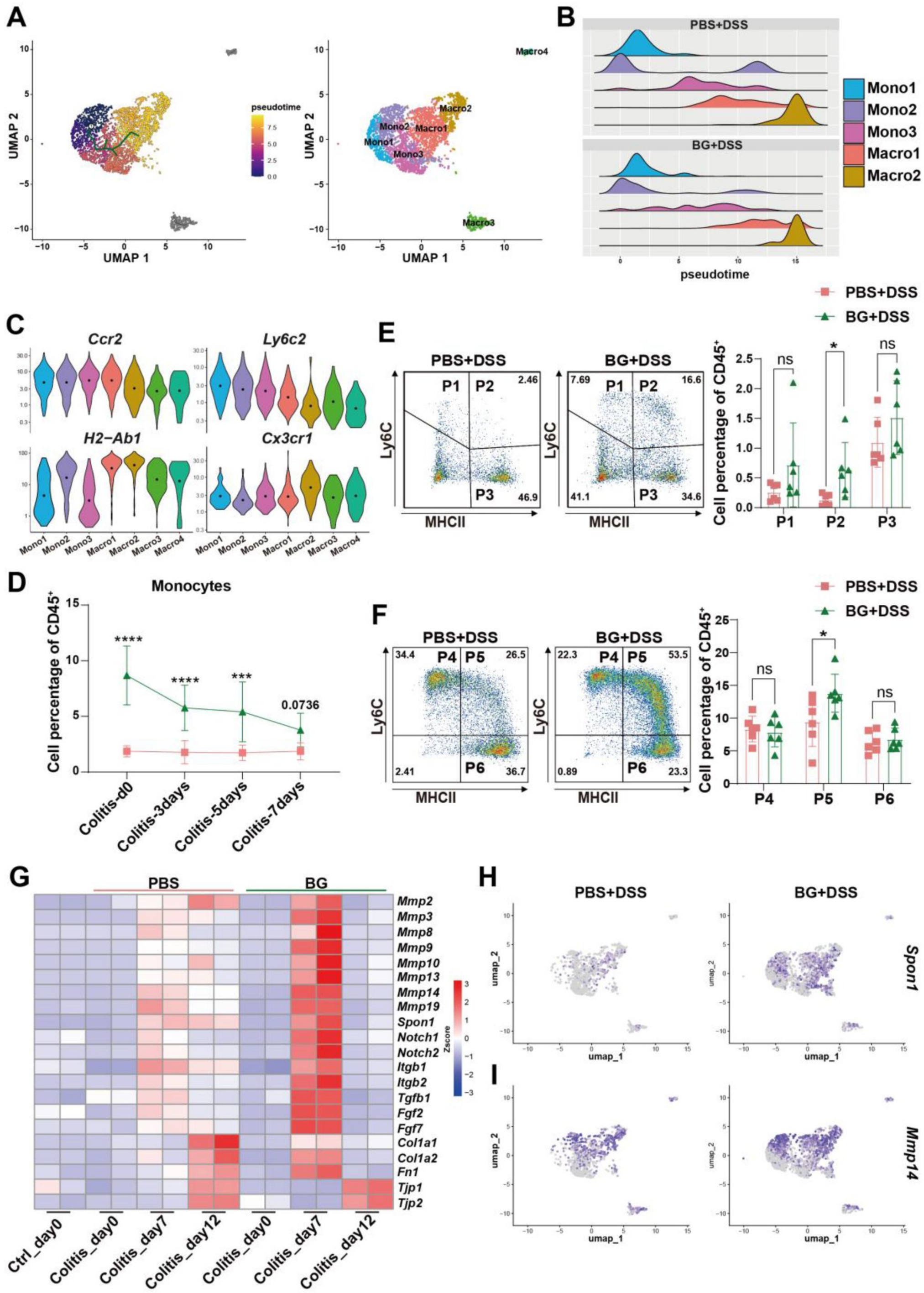


Fig 5.

**BG-mediated reprogramming of myeloid differentiation trajectories balances inflammation and enhances mucosal repair in colitis.**

(**A**) Monocle 3 trajectory analysis of monocyte/macrophage subsets, (**B**) Ridgeline plot of monocyte/macrophage subsets, (**C**) Violin plots of surface marker gene expression in monocyte/macrophage subsets, (**D**) Percentage of monocytes in peripheral blood at different time points of colitis progression, (**E and F**) Colonic LPMCs were collected, and the percentages of monocyte/macrophage were analyzed on day 7 of colitis, (**G**) Expression of mucosal repair-related genes at different time points of colitis, (**H**) Gene expression analysis of the monocyte/macrophage lineage. Data are presented as mean ± SD. Statistical significance: *$p$ <0.05, ***$p$ <0.001; ****$p$ <0.0001. ns, not significant.

# Discussion

Trained immunity, a non-antigen-specific form of innate immunological memory, enables rapid host defense mobilization upon pathogen challenge(34, 35). In inflammatory bowel disease, gut microbial dysbiosis manifests as reduced α-diversity, diminished commensal taxa (e.g., *Faecalibacterium prausnitzii*), and pathobiont expansion (e.g., *Enterobacteriaceae*, *adherent-invasive Escherichia coli*), which may compromise intestinal barrier integrity and precipitate microbial translocation(36–38). Additionally, the long-term use of immunosuppressants and biologics in IBD patients elevates infection susceptibility, rendering the microbial-controlling properties of trained immunity a viable therapeutic strategy for IBD(39). However, Excessive or sustained immune activation could trigger chronic inflammatory cascades and provoke autoimmune tissue damage through aberrant immune recognition, such as rheumatoid arthritis and cardiovascular diseases(9, 40–42). However, the functional correlation between trained immunity and colitis remains unclear, and how to apply trained immunity to colitis treatment represents an urgently unresolved scientific question.

Here, we demonstrated that BG-induced trained immunity confers cross-pathogen protective effects using a lethal *Staphylococcus aureus* infection model, while failing to induce intestinal inflammation under conditions of an intact mucosal barrier. Further investigations revealed that BG effectively attenuated dextran sulfate sodium (DSS)-induced colitis upon secondary challenge. BG pretreatment conferred long-term protection against DSS-induced colitis, with benefits lasting for at least two months after the peripheral myeloid cell populations returned to baseline levels. Bone marrow transplantation experiments confirmed that BG-induced trained immunity persists within hematopoietic progenitors, leading to durable reprogramming of peripheral monocytes. This long-term effect is likely supported by epigenetic modifications in hematopoietic cells, as reported in previous studies(21, 43). These findings highlight the potential of BG-induced trained immunity in establishing long-lasting immune memory, which could be leveraged to achieve sustained therapeutic effects in IBD.

The conventional understanding of IBD pathogenesis has long posited that abnormal immune activation is central to disease development and progression. This paradigm has predominantly focused on the role of adaptive immunity, particularly the dysregulation of Th17 and Treg functions over the past decades(44–46). However, our results suggest that BG-induced protection is independent of adaptive immune responses. Even in *Rag1*$^{-/-}$ mice, which lack adaptive immunity, BG pretreatment significantly improved DSS-induced colitis outcomes. This underscores the dominant role of innate immunity.

In DSS-induced colitis, epithelial barrier disruption facilitates microbiota translocation across the mucosal epithelium, triggering inflammatory responses mediated by innate cells(38, 47, 48). We identified monocytes as key mediators of BG-induced protection against DSS-induced colitis, as demonstrated by experiments using *Ccr2*$^{-/-}$ mouse models and monocyte transfer studies. BG pretreatment enriched Ly6C$^{hi}$CCR2$^{+}$ monocytes, which exhibited enhanced antimicrobial capabilities, as revealed by scRNA-seq. Ly6C$^{hi}$CCR2$^{+}$ monocytes displayed higher levels of antimicrobial receptors like NOD2, TLR4 and GBP-related gene. Guanylate-binding protein (GBP) enhances *Salmonella* clearance by promoting inflammasome activation and intracellular bacterial elimination, particularly through disrupting *Salmonella*-containing vacuoles and inducting pyroptosis(27, 49). Our finding revealed that BG pretreatment increased resistance to *Salmonella typhimurium* infection. As inflammation and infection is partially controlled, monocytes can differentiate into macrophage subsets with anti-inflammatory and tissue repair functions, secreting cytokines such as IL-10, TGF-β and VEGF to promote inflammation resolution and tissue repair(50, 51).

Previous studies have underscored the essential role of circulating Ly6C$^{hi}$ monocytes in replenishing intestinal macrophages(52, 53). During colitis and infection, these monocytes are rapidly recruited to injured colonic tissues, where they differentiate into Cx3cr1$^{+}$ macrophages. We also observed that these monocytes exhibited enhanced migratory capabilities, enabling them to infiltrate inflamed tissues and differentiate into CX3CR1$^{+}$ macrophages. These macrophages contribute to mucosal repair by clearing pathogens such as *Salmonella* and promoting epithelial regeneration(54). Single-cell trajectory analysis further showed that BG training promoted the differentiation of Macro1 into Macro2, which is consistent with the dynamic process of monocyte-to-mature macrophage conversion observed in *CX3CR1-GFP* reporter mice. Maintaining a balance between inflammatory monocytes, differentiated macrophages, and tissue-resident macrophages is critical for resolving inflammation, promoting tissue repair(33, 55). These findings suggest that BG not only augments monocyte antimicrobial functions but also facilitates their differentiation into macrophages, enhancing their ability to recognize, phagocytose, and eliminate bacteria while promoting tissue repair. This dual function likely strengthens the intestinal barrier, mitigates inflammation, and supports gut homeostasis.

In the clinical exploration of IBD treatment, novel therapies based on cellular regeneration (intestinal organoid transplantation) and fecal microbiota transplantation (FMT) are emerging as critical avenues to overcome the limitations of conventional pharmacotherapies(56–58). Given the efficacy of BG-induced trained immunity in alleviating DSS-induced colitis, this approach holds significant promise as a novel therapeutic strategy for IBD treatment. We transferred BG-preconditioned monocytes significantly alleviated clinical symptoms, including weight loss and colon shortening, demonstrating the potential of cell-based therapies (**Fig S10A-C**).

In conclusion, BG confers long-term colitis amelioration through trained immunity by reprogramming bone marrow-derived monocytes. Mechanistically, BG enhances monocyte chemotaxis and antimicrobial functionality activation, enabling efficient microbial containment. Crucially, this trained state establishes self-regulating circuits, engaging in negative feedback regulation of excessive immune responses, while accelerated CX3CR1$^{+}$ macrophage differentiation facilitates mucosal repair. These findings establish a foundation for leveraging trained immunity as a therapeutic strategy for IBD, highlighting the potential of BG in modulating intestinal immunity and restoring gut homeostasis.

# Materials and methods

## Animals

C57BL/6J wild-type male mice (6-8weeks) were purchased from Xiamen University Laboratory Animal Center. *Rag1*$^{-/-}$ and *CX3CR1-GFP* mice were generously provided by Dr. Kairui Mao and Dr.Xiaofen Chen from Xiamen University, respectively. *Ccr2*$^{-/-}$ mice were kindly provided by Dr. Shih-Chin Cheng from Xiamen University. Mice were maintained under specific pathogen-free conditions. All animal protocols were reviewed and approved by the Institutional Animal Care and Use Committee of Xiamen University.

## In vivo beta-glucan (BG) training and DSS colitis model

Mice were i.p. injected with one dose of BG (1 mg) on day -7/-49, Then, Mice received 3% indicated DSS (36,000 to 50,000 molecular weight; MP Biomedicals) in their drinking water for 5 days, followed by 2 days or 7 days of distilled water without DSS. Control animals received distilled water for the entire period. Mice were monitored every day from day 0 for body weight until day 7 or 12.

## Histology

Colons were collected from sacrificed mice at the end of each treatment schedule, as described above, colon was fixed with 4% paraformaldehyde and embedded in Paraffin. Tissue sections (5 μm) were prepared, deparaffinized, and stained with hematoxylin and eosin. with minor modifications, by assessing mucosa thickening, inflammatory cells, and submucosa cell infiltration. Each criterion was scored as 0-4, and the sum of each score was defined as the histological score. Histological scores were assigned by experimenters "blinded" to sample identity.

## Coloscopy

A coloscopy system (ENDOCAM® Logic HD, RICHARD WOLF) was used to monitor the severity of DSS-induced colitis. After an 8-hour fast, we administered isoflurane (RWD, Co. Ltd.) and applied glycerin to the anus for smooth insertion.

## In vivo intestine permeability assay

Age and sex-matched mice were orally administered with 0.6 mg/g body weight of an 100 mg/ml solution of FITC-dextran (FD4, Sigma). 5 hours later, retro-orbital blood was collected from each mouse. Serum was prepared by allowing the blood to clot by leaving it undisturbed overnight at 4°C and then subsequently centrifuged at 5000 rpm for 10 minutes. Dilutions of FITC-dextran in PBS and separately in pooled mouse serum were used as a standard curve. Absorbance of 50 μl serum was measured at microplate reader with excitation and emission filters set at 490 and 530 nm, respectively. Experiments were performed at least two independent times each in triplicate.

## RNA extraction and real-time PCR

Total RNA was extracted from colon tissue previously washed from luminal content. Samples were homogenized in TRIZOL. Homogenized tissues were then added with chloroform in order to separate RNA from genomic DNA and proteins. RNA isolation and purification required isopropanol (RNA precipitation) and 70% ethanol (RNA wash). Total RNA was resuspended in Nuclease-free water. Subsequently, 1 μg of RNA was reverse-transcribed using the Hifair® ll lst Strand cDNA Synthesis SuperMix for qPCR (gDNA digester plus) following manufacturer's instructions. cDNA analysed through quantitative Real-Time PCR.

## Isolation of cells from lamina propria

To isolate lamina propria mononuclear cells (lamina propria MCs), extraintestinal fat tissue was carefully removed and colons were then flushed of their luminal content with physiologic solution, opened longitudinally and cut into 1 cm pieces. Epithelial cells and mucus were removed by incubation with 10 mL of D-Hank's balanced salt solution (containing EDTA and DTT, free of $Ca2^{+}$ and $Mg2^{+}$) for 20 minutes at 37°C at 200 rpm in a constant temperature shaker. Colon pieces were then digested in Hank's (with $Ca2^{+}$ and $Mg2^{+}$) containing 0.5 mg/ml Collagenase IV and 15 μg/ml DNase I, for 40 min at 37°C shaking at 200 rpm. The remaining cells were centrifuged and resuspended in FACS buffer (1% FBS, 2 mM EDTA in PBS).

## Bone Marrow Transplantation

Wild type CD45.2 mice were subjected to 8 Gy irradiation (RAD SOURCE, RS2000). CD45.1 wild type and BG trained mice were sacrificed, and the femurs were used to harvest bone marrow cells. $1\times10^{7}$ bone marrow cells from CD45.1 wild type and BG trained donor mice were transferred to irradiated CD45.2 recipients via tail-vein injection. 2.5% DSS was given at 6 weeks-post bone marrow transplantation. Mice were monitored every day until day 12.

## Adoptive transfer of bone marrow monocytes

BG trained mice were sacrificed, and the femurs were used to harvest bone marrow cells, Bone marrow cells were stained with the following panel: Biotin antibody (B220, CD4, CD8, NK1.1, Ly6G), Live/dead (Cat. L34976, Invitrogen), CD45 (Cat. 45-0451-82, Clone. 30-F11, ebioscience), CD11b (Cat. 48-0112-82, Clone. M1/70, ebioscience), Ly6C (Cat. 128016, Clone. HK1.4, Biolegend). Biotin-labeled lineage antibodies were used to stain bone marrow cells at 4℃ for 20 min and then incubated with fluorescence conjugated streptomycin together with other surface protein antibodies for another 20 min. Ly6C$^{hi}$ monocytes were sorted to > 97% purity using a BD Aria III cell sorter.

## RNA-seq

RNA was extracted from colon LPLs of individual mice. RNA integrity was assessed using the RNA Nano 6000 Assay Kit of the Bioanalyzer 2100 system (Agilent Technologies, CA, USA). The clustering of the index-coded samples was performed on a cBot Cluster Generation System using TruSeq PE Cluster Kit v3-cBot-HS (Illumia) according to the manufacturer's instructions. The library was sequenced on an Illumina Novaseq platform and 150 bp paired-end reads were generated.

## scRNA-seq

Isolated lamina propria mononuclear cells, sample from three mice were pooled together and analyzed per condition. For scRNA-seq of Colon CD45$^+$ cells were sorted by BD Arial III. These cells were barcoded with 10×Cellplex oligos before being encapsulated using the 10X Chromium 3' Reagent Kits v3 according to manufacturer's instructions.

## RNA-seq data processing and analysis

Raw data of fastq format were quality checked through fastqc software. Reference genome and gene model annotation files were downloaded from genome (mm10) website directly. Index of the reference genome was built using Hisat2 v2.1.0 and paired-end clean reads were aligned to the reference genome using Hisat2 v2.1.0. Expression levels across the sample were expressed in FPKM (Fragment per kilobase per million). Weighted correlation network analysis was perform using an R package from horvath.genetics.ucla.edu/html/CoexpressionNetwork/Rpackages/WGCNA/. Differential gene expression was used DESeq2 by a twofold cutoff and pvalue < 0.05. All differentially expressed genes were plotted with the "pheatmap" and "ggplot2" R packages. For KEGG enrichment analysis, p-value < 0.05 was used as a threshold to determine significant enrichment for gene sets with the "clusterProfiler" R package.

## scRNA-seq data processing and analysis

Raw sequencing data was processed and aligned mm10 mouse reference genome with CellRanger (10×Genomics) v 7.2.0. Resulting filtrated matrices (count matrices) of molecular counts were used as input for further processing with Seurat package V5.0.1running under R studio. First, quality control was performed to create Seurat object with min features > 200 and removal of cells having < 200 or > 8000 expressed genes or > 5% mitochondrial counts. The total number of recovered cells was mentioned before. Variable features using FindVariableFeatures (using RNA and vst as an assay and selection method as parameters) and normalization using normalization.method = "LogNormalize", scale.factor = 10000 were performed . This integrated data is scaled using ScaleData function using all genes and "RNA" as assay method.

Doublet cells were filtered by DoubletFinder v3. Principal Component Analysis was performed on variable features using RunPCA and first 30 PCs were chosen. The nearest neighbors using the FindNeighbors(dims=1:12) and FindClusters(resolution=0.8) function. For visualization, the nonlinear dimensional reduction such as UMAP analysis was using RunUMAP function from Seurat. To calculate pseudotime based on the Mono/Marco subset data using the monocle3.

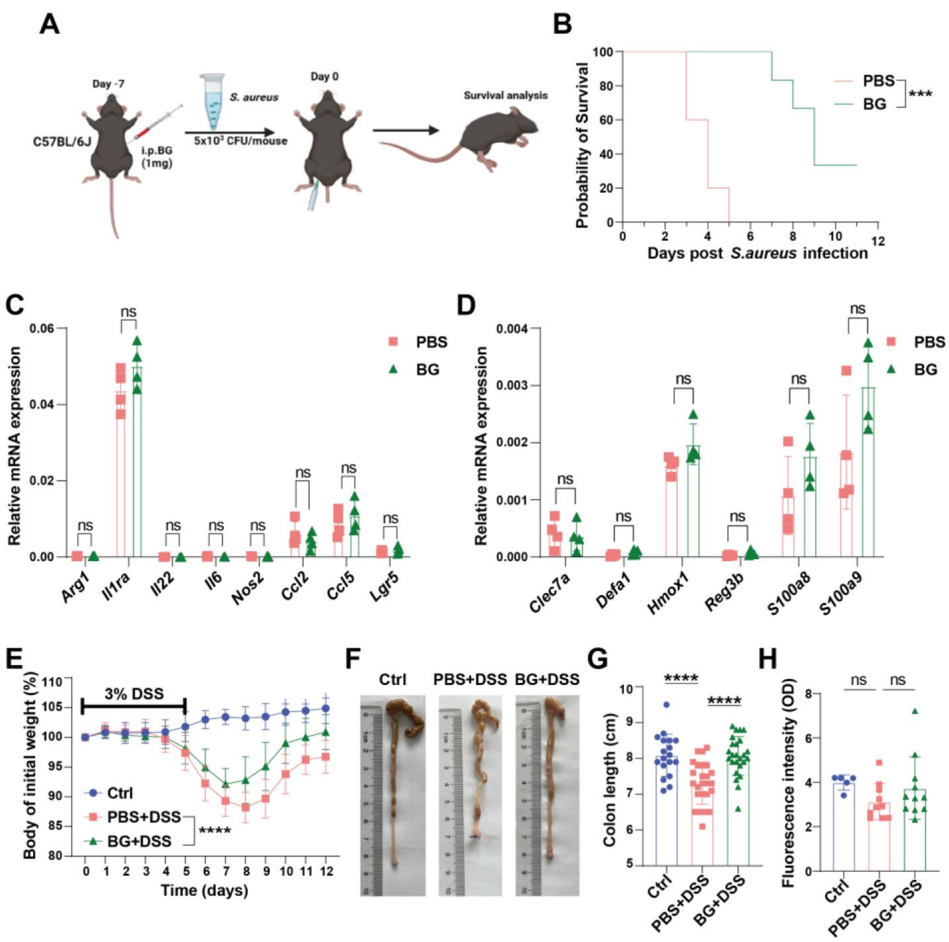
### Statistical Analysis

Statistical analyses were conducted using GraphPad Prism version 9.0 and R statistical software (version 4.2.2, R Foundation for Statistical Computing). Data were analyzed using a two-tailed Student's *t*-test. Multiple-group comparisons were performed using one-way or two-way analysis of variance (ANOVA) followed by Tukey's multiple-comparisons test. Survival data were assessed using Kaplan-Meier survival plots, followed by the log-rank test. Significant differences were indicated by an asterisk ($p < 0.05$).

## Data availability statement

All data relevant to the study are included in the manuscript or uploaded as online supplemental information. Sequencing data have been deposited to the GEO with accession number GSE285859 (single cell RNA-seq), GSE285860 (RNA-seq).

## Supplementary figures




## Fig S1.

### BG pretreatment ameliorates DSS-induced colitis.

(**A**) Schematic representation of BG-induced trained immunity and *Staphylococcus aureus* infection model, (**B**) Survival curve of mice infected with *Staphylococcus aureus,* (**C and D**) The gene expression of inflammatory mediators and antibacterial were analyzed by qRT-PCR, (**E**) Body weight change curve of mice pretreated with BG for one week, followed by colitis induction with 3% DSS ($n$ = 18-25), (**F and G**) Changes of colon length from (**E**), (**H**) FITC-dextran assay assessing intestinal barrier function ($n$ = 5-11). Data are presented as mean ± SD. Statistical significance: *$p$ < 0.05, ***$p$ < 0.001; ****$p$ < 0.0001. ns, not significant.

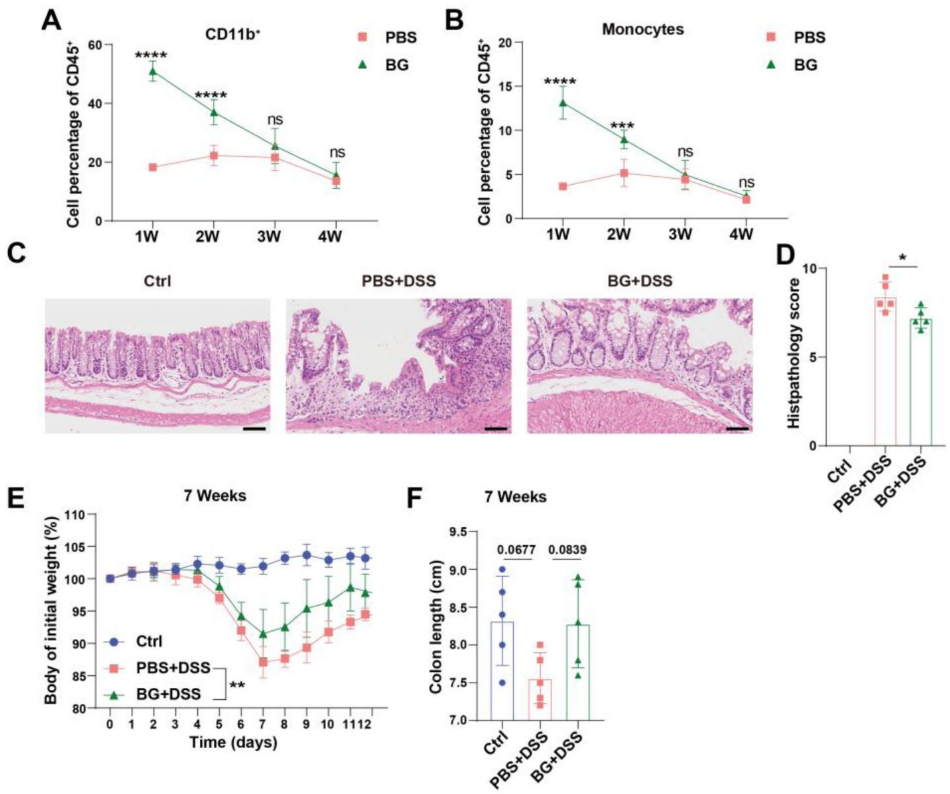



**Fig S2.**

### BG pretreatment 4 or 7 weeks ameliorates DSS-induced colitis.

(**A and B**) Percentage of myeloid cells and monocytes in the peripheral blood of mice at different time points after one week of BG pretreatment, (**C and D**) H&E staining, and histological scoring of colitis mice after four weeks of BG training. Scale bars: 100 μm, (**E**) Body weight change curve of mice pretreated with BG for seven weeks, followed by colitis induction with 3% DSS, (**F**) Changes of colon length from (**E**) ($n$ = 5). Data are presented as mean ± SD. Statistical significance: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$. ns, not significant.

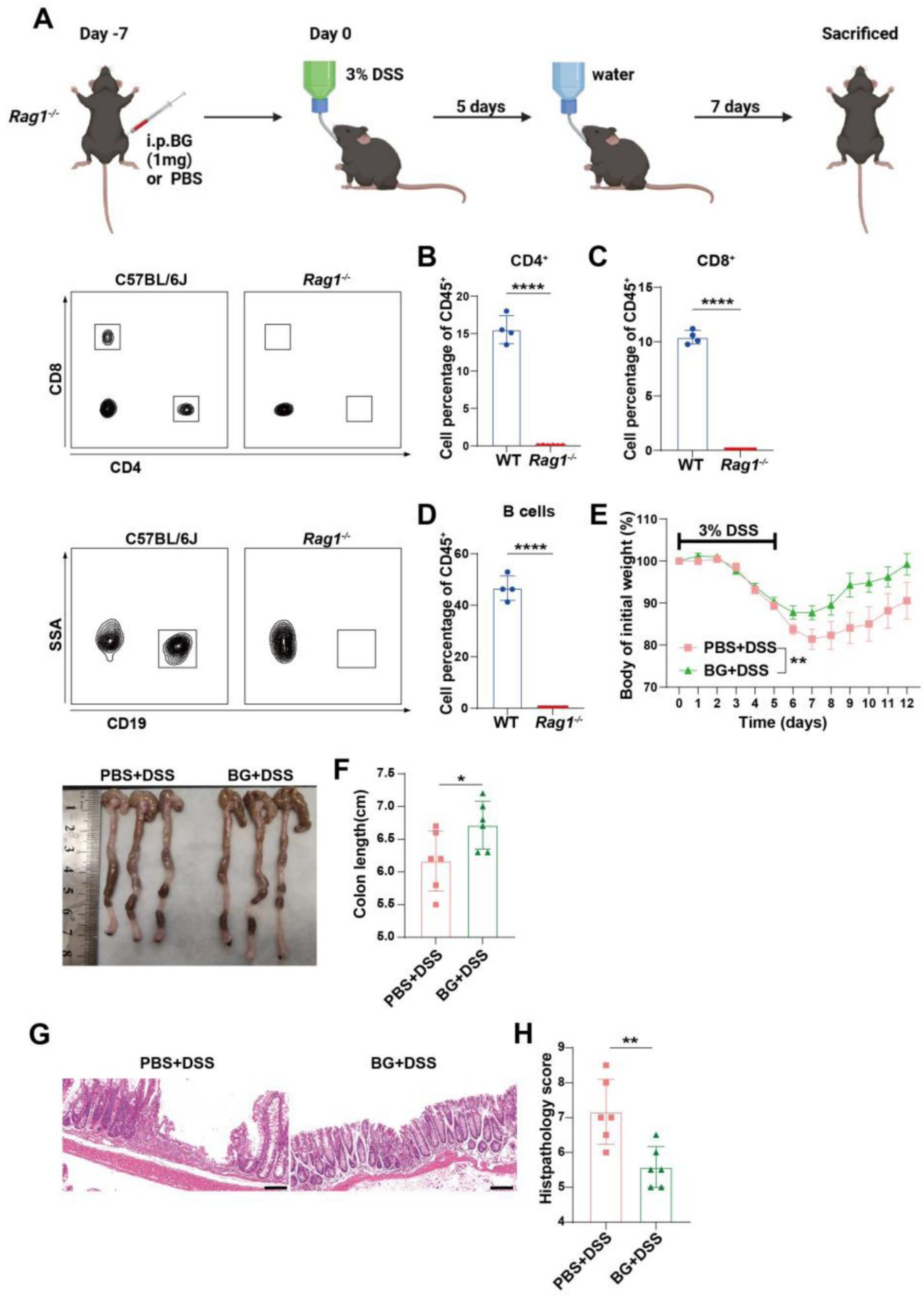
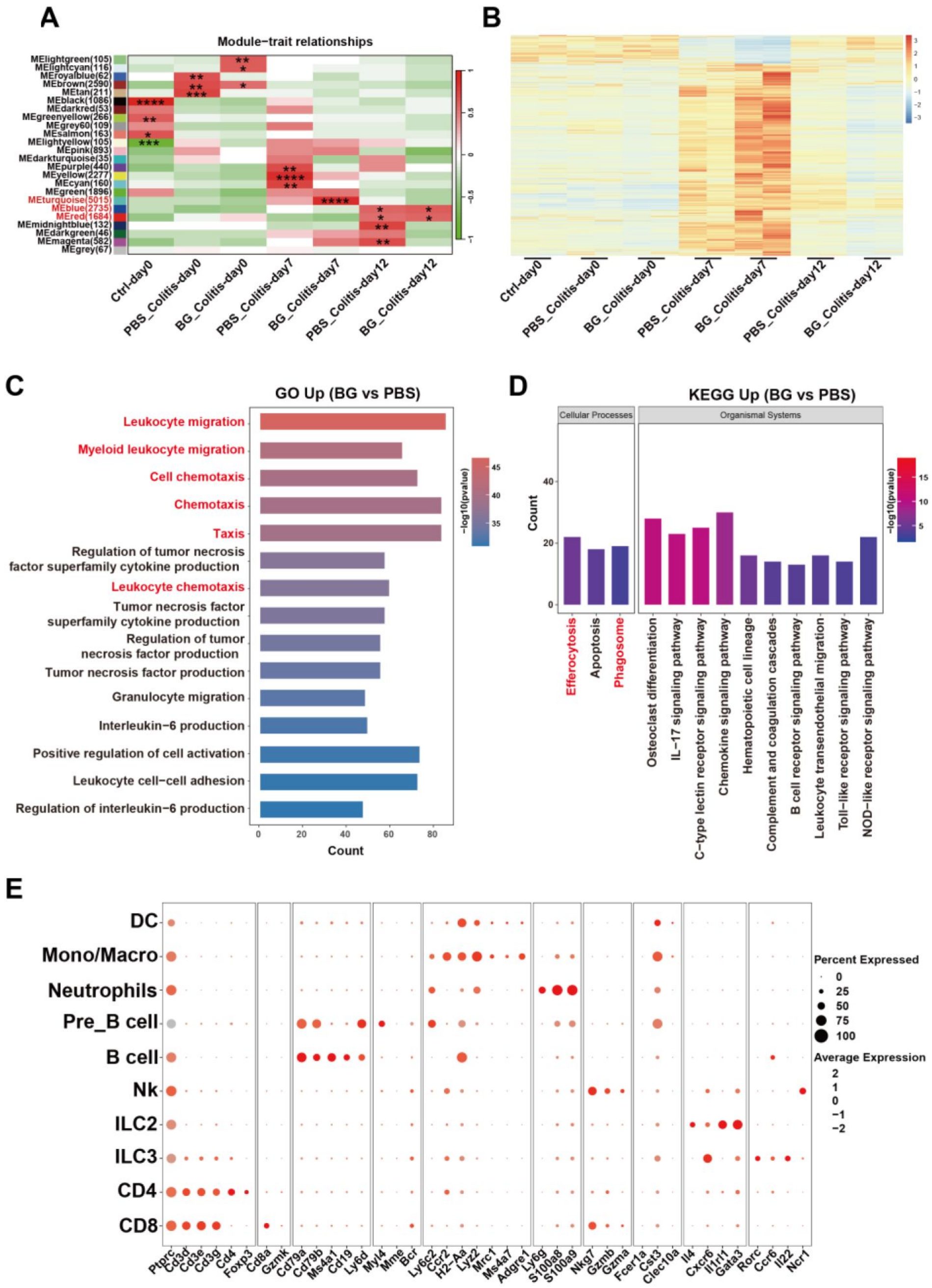



**Fig S3.**

## BG pretreatment ameliorates colitis is independent of adaptive immunity.

(**A**) Schematic representation of BG-induced trained immunity and DSS colitis model in *Rag1*$^{-/-}$ mice, (**B-D**) Flow cytometry analysis of peripheral blood from *Rag1*$^{-/-}$ mice, (**E**) Body weight change curve of *Rag1*$^{-/-}$ mice pretreated with BG for one week, followed by colitis induction with 3% DSS, (**F**) Colon length changes in *Rag1*$^{-/-}$ mice from (**E**), (**G and H**) H&E staining and histological scoring. Scale bars: 100 μm. Data are presented as mean ±SD. Statistical significance: *$p$ < 0.05, **$p$ < 0.01, ****$p$ < 0.0001.




## Fig S4.

### BG ameliorates colitis by enhancing myeloid cell activation.

(**A**) RNA sequencing of colon tissue at different time points during colitis, with WGCNA identifying major gene modules, (**B**) DEGs in the Meturquoise module, (**C and D**) GO and KEGG pathway analysis of colon tissue on day seven of colitis after one week of BG pretreatment, (**E**) Expression of marker genes in LP CD45$^+$ cells on day seven of colitis after one week of BG pretreatment.

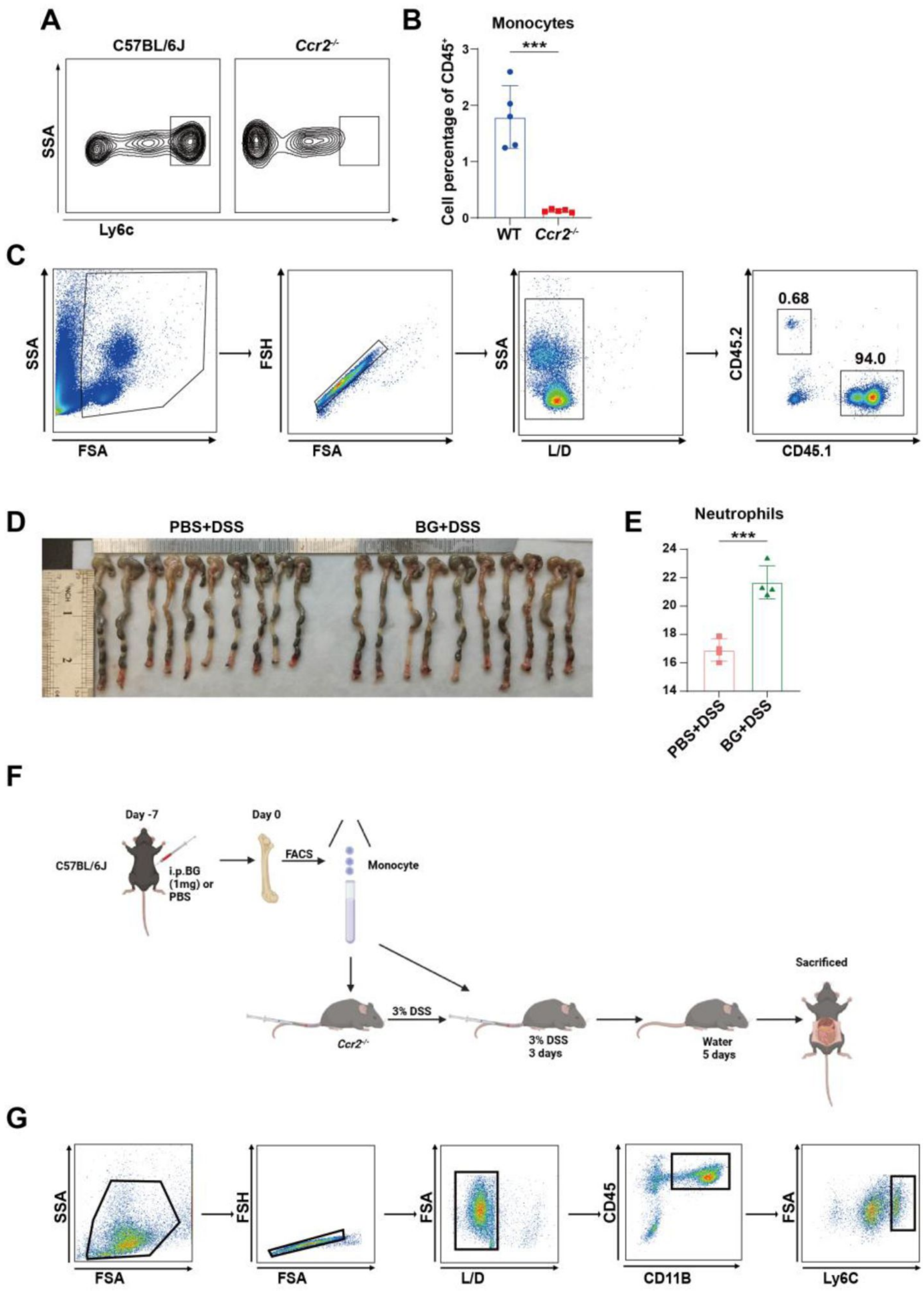



**Fig S5.**

### BG trained bone marrow monocytes protected against colitis via the Ccl2-Ccr2 axis.

(**A and B**) Flow cytometry analysis of peripheral blood from *Ccr2*$^{-/-}$mice, (**C**) Flow cytometry analysis of bone marrow reconstitution in CD45.2 recipient mice, (**D**) Colon length changes in bone marrow transplantation experiments from BG-pretreated CD45.1 mice to CD45.2 mice, (**E**) Percentage of Ly6G$^{+}$ neutrophils in peripheral blood of CD45.2 recipient mice, (**F**) Bone marrow monocytes transplant model, (**G**) Flow sorting scheme of bone marrow monocytes. Data are presented as mean ±SD. Statistical significance: ***$p$ < 0.001.

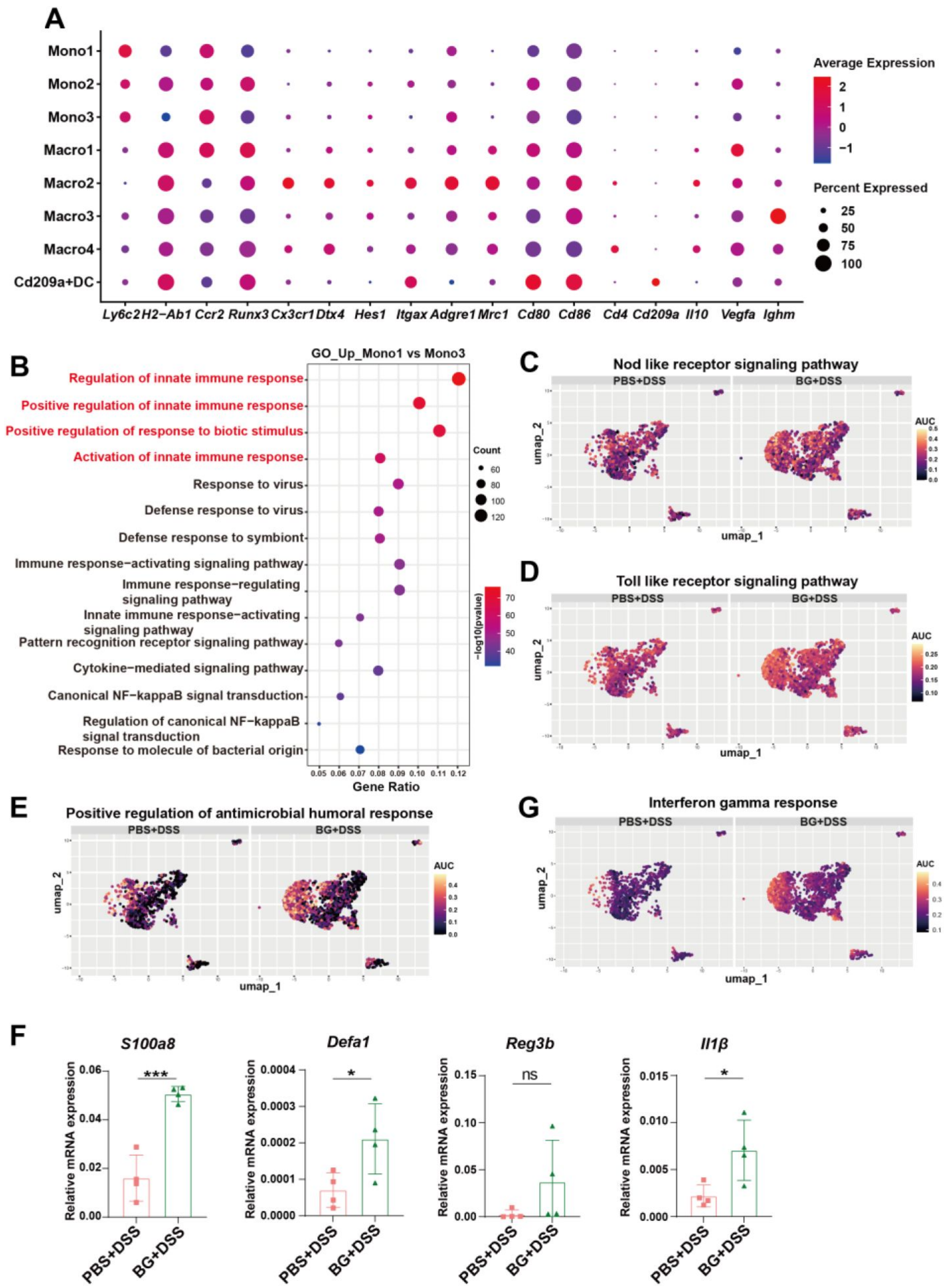



**Fig S6.**

### BG pretreatment enhances innate immunity and phagocytic capacity.

(**A**) Major marker genes of eight cell clusters within the monocyte/macrophage lineage, (**B**) GO analysis of genes upregulated in monocytes 1 and monocyte 3, (**C-E**) AUC scores of selected pathways, (**F**) Antibacterial gene expression was analyzed by qRT-PCR. (**G**) AUC scores of selected pathways. Data are presented as mean ± SD. Statistical significance: *$p$ < 0.05, ***$p$ < 0.001, ns, not significant.

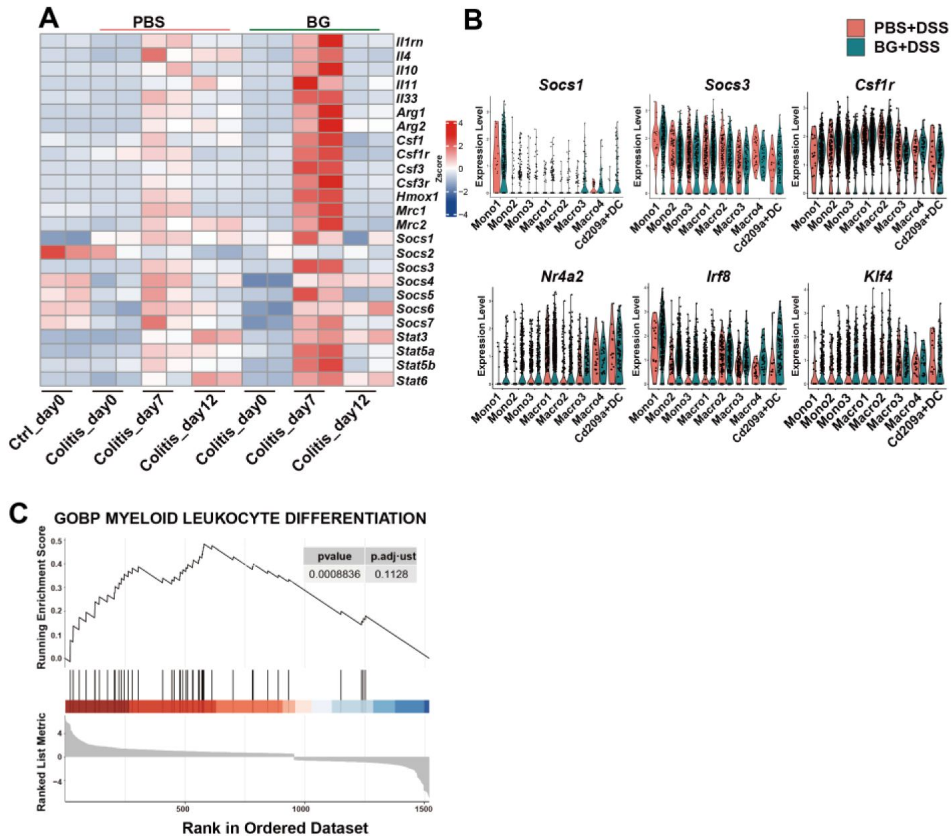
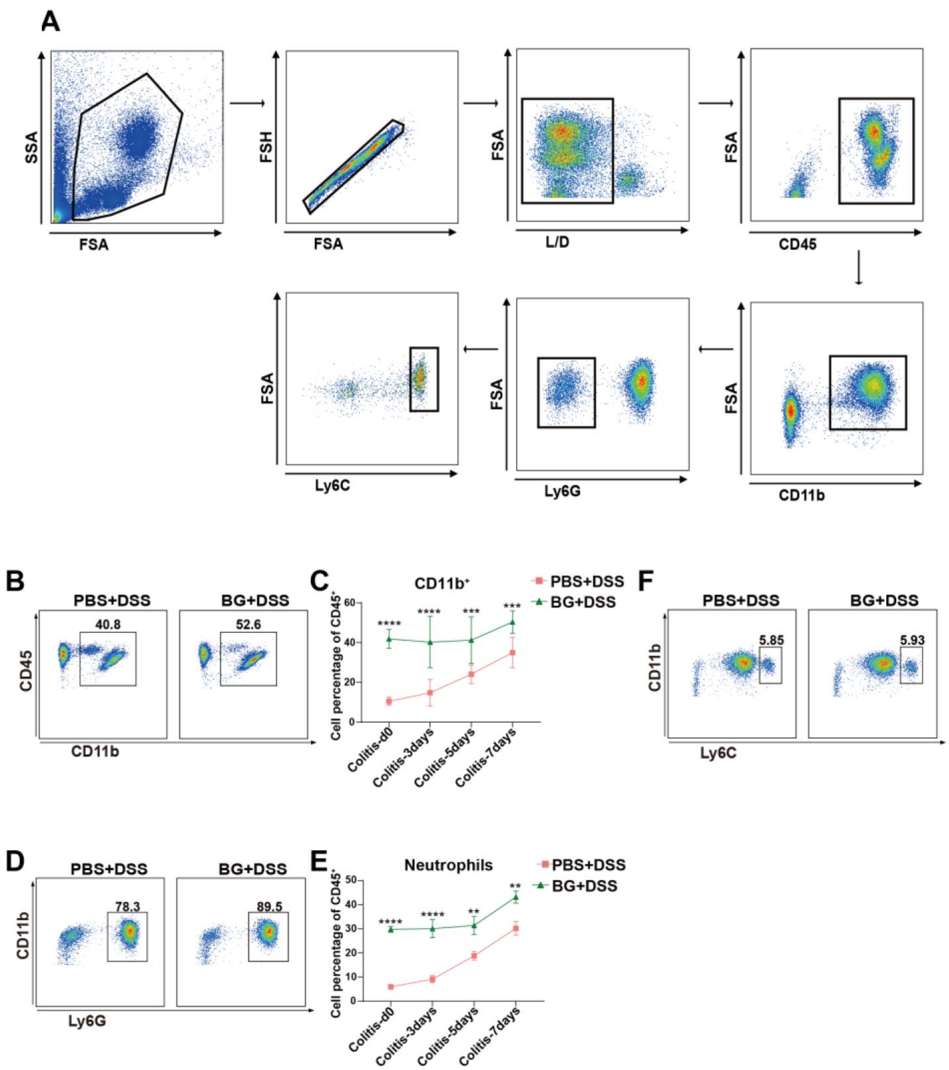



**Fig S7.**

### BG-induced trained immunity promotes monocyte differentiation.

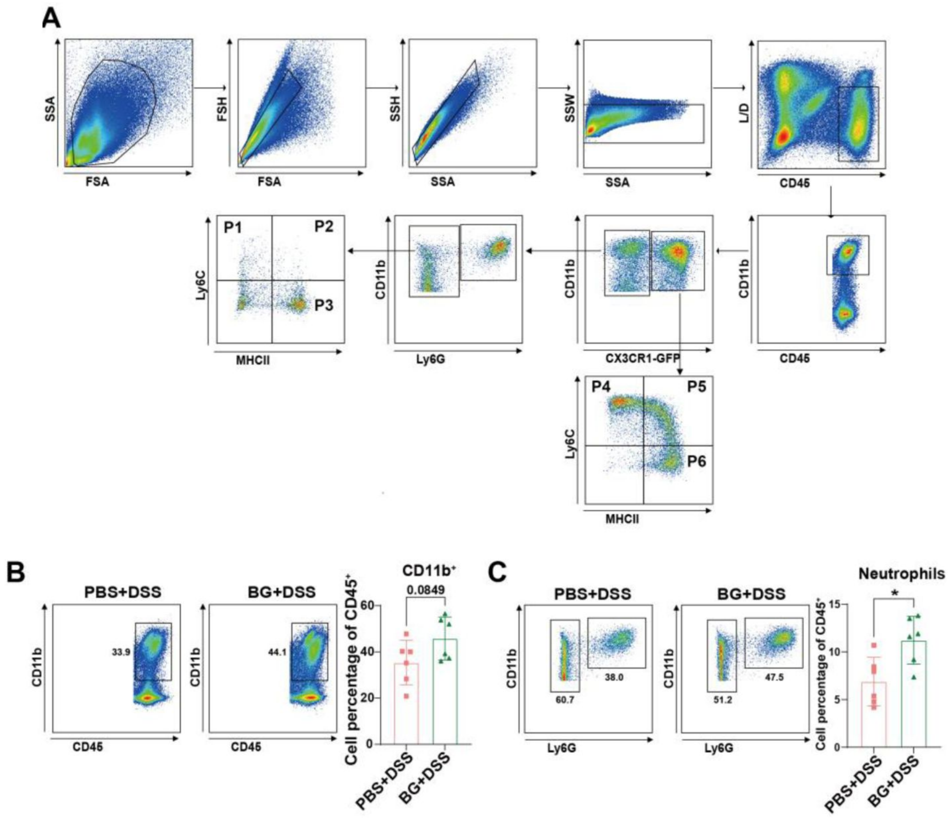
(**A**) Gene expression analysis at different time points of colitis, based on colon RNA sequencing, (**B**) Violin plots of gene expression in eight cell clusters, (**C**) GSEA analysis showing gene enrichment pattern on day seven of colitis.




## Fig S8

### BG training downregulates the proportion of peripheral monocytes during colitis development.

(**A**) Peripheral blood flow cytometry gating strategy, (**B**) Representative flow cytometry plots of CD11b⁺ myeloid cells in peripheral blood on day seven of colitis (**C**) The percentage of $CD11b^+$ myeloid cells at different time points of colitis, (**D**) Representative flow cytometry plots of $Ly6G^+$ neutrophils in peripheral blood on day seven of colitis, (**E**) Percentage of $Ly6G^+$ neutrophils at different time points of colitis, (**F**) Representative flow cytometry plots of $Ly6C^{hi}$ monocytes in peripheral blood on day seven of colitis. Data are presented as mean ±SD. Statistical significance: $^{**}p < 0.01$; $^{***}p < 0.001$; $^{****}p < 0.0001$.




## Fig S9

### BG training upregulates the proportion of colonic monocytes/macrophages in colitis mice.

(**A**) Colonic LPMCs flow cytometry gating strategy, (**B and C**) The percentages of CD11b$^+$ myeloid cells (**B**) and Ly6G$^+$ neutrophils (**C**) were analyzed on day seven of colitis ($n$ = 6). Data are presented as mean ±SD. Statistical significance: *$p$ < 0.05. ns, not significant.

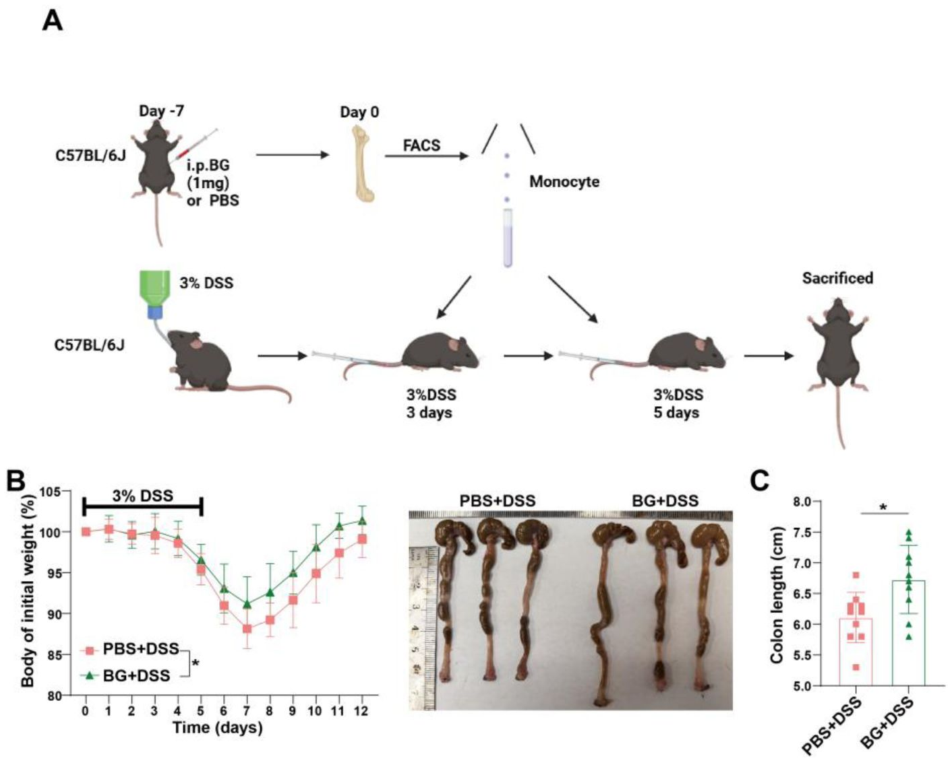



**Fig S10**

### Adoptive transfer of BG-trained monocytes ameliorates experimental colitis.

(**A**) Bone marrow monocytes transplant and colitis treatment model, (**B and C**) Body weight change curve (**B**) and colon length changes (**C**) in WT colitis mice receiving Ly6C$^{hi}$ monocyte adoptive transfer ($n$ = 10). Data are presented as mean ±SD. Statistical significance: *$p$ < 0.05.

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## Additional information

### Author contributions

Y.Y.L. designed and did the experiments, analyzed the data and wrote the paper; Y.Y.F., Q.X.G. and Q.Y. C. analyzed the data, wrote the paper, and supervised the study. Y.Q.H., L.W., H.X.S., E.M.C., Q.Y.X., Y.C., Q.Q.F., L.L.L provided advice and oversaw a portion of the work. J.L.R., S.-C.C., H.Z.X. conceived the idea, secured funding, supervised the study, helped in data interpretations, revised drafts of manuscript; supervised overall project. All authors reviewed and approved the final version of the manuscript.

### Ethics approval

All animal protocols were reviewed and approved by the Institutional Animal Care and Use Committee of Xiamen University.

Patient consent for publication: Not applicable.


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**Reviewer #1 (Public review):**

Summary:

This study presents an interesting investigation into the role of trained immunity in inflammatory bowel disease, demonstrating that β-glucan-induced reprogramming of innate immune cells can ameliorate experimental colitis. The findings are novel and clinically relevant, with potential implications for therapeutic strategies in IBD. The combination of functional assays, adoptive transfer experiments, and single-cell RNA sequencing provides comprehensive mechanistic insights. However, some aspects of the study could benefit from further clarification to strengthen the conclusions.

Strengths:

(1) This study elegantly connects trained immunity with IBD, demonstrating how β-glucan-induced innate immune reprogramming can mitigate chronic inflammation.

(2) Adoptive transfer experiments robustly confirm the protective role of monocytes/macrophages in colitis resolution.

(3) Single-cell RNA sequencing provides mechanistic depth, revealing the expansion of reparative $Cx3cr1^+$ macrophages and their contribution to epithelial repair.

(4) The work highlights the therapeutic potential of trained immunity in restoring gut homeostasis, offering new directions for IBD treatment.

Weaknesses:

While β-glucan may exert its training effect on hematopoietic stem cells, performing ATAC-seq on HSCs or monocytes to profile chromatin accessibility at antibacterial defense and mucosal repair-related genes would further validate the trained immunity mechanism. Alternatively, the authors could acknowledge this as a study limitation and future research direction.

https://doi.org/10.7554/eLife.107339.1.sa2

**Reviewer #2 (Public review):**

Summary:

The study investigates whether β-glucan (BG) can reprogram the innate immune system to protect against intestinal inflammation. The authors show that mice pretreated with BG prior to DSS-induced colitis experience reduced colitis severity, including less weight loss, colon damage, improved gut repair, and lowered inflammation. These effects were independent of adaptive immunity and were linked to changes in monocyte function.

The authors show that the BG-trained monocytes not only help control inflammation but confer non-specific protection against experimental infections (Salmonella), suggesting the involvement of trained immunity (TI) mechanisms. Using single-cell RNA sequencing, they map the transcriptional changes in these cells and show enhanced differentiation of monocytes into reparative $CX3CR1^+$ macrophages. Importantly, these protective effects were transferable to other mice via adoptive cell transfer and bone marrow transplantation, suggesting that the innate immune system had been reprogrammed at the level of stem/progenitor cells.

Overall, this study provides evidence that TI, often associated with heightened inflammatory programs, can also promote tissue repair and resolution of inflammation. Moreover, this BG-induced functional reprogramming can be further harnessed to treat chronic inflammatory disorders like IBD.

Strengths:

(1) The authors use advanced experimental approaches to explore the potential therapeutic use of myeloid reprogramming by β-glucan in IBD.

(2) The authors follow a data-to-function approach, integrating bulk and single-cell RNA sequencing with in vivo functional validation to support their conclusions.

(3) The study adds to the growing evidence that TI is not a singular pro-inflammatory program, but can adopt distinct functional states, including anti-inflammatory and reparative phenotypes, depending on the context.

Weaknesses:

(1) The epigenetic and metabolic basis of TI is not explored, which weakens the mechanistic claim of TI. This is especially relevant given that a novel reparative, anti-inflammatory TI program is proposed.

(2) The absence of a BG-only group limits interpretation of the results. Since the authors report tissue-level effects such as enhanced mucosal repair and transcriptional shifts in intestinal macrophages (colonic RNA-Seq), it is important to rule out whether BG alone could influence the gut independently of DSS-induced inflammation.
Without a BG-only control, it is hard to distinguish a true trained response from a potential modulation caused directly by BG.

(3) Although monocyte transfer experiments show protection in colitis, the fate of the transferred cells is not described (e.g., homing or differentiation into $Cx3cr1^+$ macrophage subsets). This weakens the link between specific monocyte subsets and the observed phenotype.

(3) While scRNA-seq reveals distinct monocyte/macrophage subclusters (Mono1-3..), their specific functional roles remain speculative. The authors assign reparative or antimicrobial functions based on transcriptional signatures, but do not perform causal experiments (depletion or in vitro assays). The biological roles of these cells remain correlative.

(4) While $Rag1^{-/-}$ mice were used to rule out adaptive immunity, the potential role of innate lymphoid cells (ILCs), particularly ILC2s and ILC3s, which are known to promote mucosal repair (PMID: 27484190), was not explored. Given the reparative phenotype observed, the contribution of ILCs remains a confounding factor.

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**Reviewer #3 (Public review):**

Summary:

In the present work, Yinyin Lv et al offer evidence for the therapeutic potential of trained immunity in the context of inflammatory bowel disease (IBD). Prior research has demonstrated that innate cells pre-treated (trained) with β-glucan show an enhanced pro-inflammatory response upon a second challenge.

While an increased immune response can be beneficial and protect against bacterial infections, there is also the risk that it will worsen symptoms in various inflammatory disorders. In the present study, the authors show that mice preconditioned with β-glucan have enhanced resistance to Staphylococcus aureus infection, indicating heightened immune responses.

The authors demonstrate that β-glucan training of bone marrow hematopoietic progenitors and peripheral monocytes mitigates the pro-inflammatory effects of colitis, with protection extending to naïve recipients of the trained cells.

Using a dextran sulfate sodium (DSS)-induced model of colitis, β-glucan pre-treatment significantly dampens disease severity. Importantly, the use of Rag1^-/- mice, which lack adaptive immune cells, confirms that the protective effects of β-glucan are mediated by innate immune mechanisms. Further, experiments using Ccr2^-/- mice underline the necessity of monocyte recruitment in mediating this protection, highlighting CCR2 as a key factor in the mobilization of β-glucan-trained monocytes to inflamed tissues. Transcriptomic profiling reveals that β-glucan training upregulates genes associated with pattern recognition, antimicrobial defense, immunomodulation, and interferon signaling pathways, suggesting broad functional reprogramming of the innate immune compartment. In addition, β-glucan training induces a distinct monocyte subpopulation with enhanced activation and phagocytic capacity. These monocytes exhibit an increased ability to infiltrate inflamed colonic tissue and differentiate into macrophages, marked by increased expression of Cx3cr1. Moreover, among these trained monocyte and macrophage subsets, other gene expression signatures are associated with tissue and mucosal repair, suggesting a role in promoting resolution and regeneration following inflammatory insult.

Strengths:

(1) Overall, the authors present a mechanistically insightful investigation that advances our understanding of trained immunity in IBD.

(2) By employing a range of well-characterized murine models, the authors investigate specific mechanisms involved in the effects of β-glucan training.

(3) Furthermore, the study provides functional evidence that the protection conferred by the trained cells persists within the hematopoietic progenitors and can be transferred to naïve recipients. The integration of transcriptomic profiling allows the identification of changes in key genes and molecular pathways underlying the trained immune phenotype.

(4) This is an important study that demonstrates that β-glucan-trained innate cells confer protection against colitis and promote mucosal repair, and these findings underscore the potential of harnessing innate immune memory as a therapeutic approach for chronic inflammatory diseases.

Weaknesses:

However, FPKM is not ideal for between-sample comparisons due to its within-sample normalization approach. Best practices recommend using raw counts (with DESeq2) for more robust statistical inference.

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**Author response:**

> ***Public Reviews:***
>
> ***Reviewer #1 (Public review):***
>
> *Summary:*
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> *This study presents an interesting investigation into the role of trained immunity in inflammatory bowel disease, demonstrating that β-glucan-induced reprogramming of innate immune cells can ameliorate experimental colitis. The findings are novel and clinically relevant, with potential implications for therapeutic strategies in IBD. The combination of functional assays, adoptive transfer experiments, and single-cell RNA sequencing provides comprehensive mechanistic insights. However, some aspects of the study could benefit from further clarification to strengthen the conclusions.*

We are grateful for the reviewer's positive assessment of our study and constructive suggestions to improve the manuscript.

> *Strengths:*
>
> *(1) This study elegantly connects trained immunity with IBD, demonstrating how β-glucan-induced innate immune reprogramming can mitigate chronic inflammation.*
>
> *(2) Adoptive transfer experiments robustly confirm the protective role of monocytes/macrophages in colitis resolution.*
>
> *(3) Single-cell RNA sequencing provides mechanistic depth, revealing the expansion of reparative Cx3cr1$^{+}$ macrophages and their contribution to epithelial repair.*
>
> *(4) The work highlights the therapeutic potential of trained immunity in restoring gut homeostasis, offering new directions for IBD treatment.*
>
> *Weaknesses:*
>
> *While β-glucan may exert its training effect on hematopoietic stem cells, performing ATAC-seq on HSCs or monocytes to profile chromatin accessibility at antibacterial defense and mucosal repair-related genes would further validate the trained immunity mechanism. Alternatively, the authors could acknowledge this as a study limitation and future research direction.*

We agree that further epigenetic profiling—such as ATAC-seq analysis on HSCs or monocytes—would provide additional mechanistic depth to our current findings. We will acknowledge this as a limitation of the present study and highlight it as an important direction for future research.

> *Comment (1): It's better to include a schematic summarizing the proposed mechanism for reader clarity.*

We agree that a visual summary will enhance the clarity and accessibility of our findings. We will add a new schematic diagram (Figure 6) illustrating the proposed mechanism of β-glucan–induced myeloid reprogramming and its protective effects in the experimental colitis model.

> *Comment (2): Discuss potential off-target effects of β-glucan-induced trained immunity (e.g., risk of exacerbated inflammation in other contexts).*

We appreciate this important comment regarding the potential off-target effects of β-glucan pretreatment. As trained immunity is known to amplify inflammatory responses upon heterologous stimulation and has been implicated in chronic inflammation–prone conditions such as atherosclerosis, this is an important consideration. Previous in vivo studies have shown that β-glucan pretreatment can enhance antibacterial or antitumor responses without inducing basal inflammation after one week of administration (PMID: 22901542, PMID: 30380404, PMID: 36604547, PMID: 33125892). Nevertheless, it remains possible that β-glucan–induced trained immunity could have unintended effects in certain contexts, which warrants further investigation and caution. We will expand the Discussion section to include a dedicated paragraph addressing these potential off-target effects.

> ***Reviewer #2 (Public review):***
>
> *Summary:*
>
> *The study investigates whether β-glucan (BG) can reprogram the innate immune system to protect against intestinal inflammation. The authors show that mice pretreated with BG prior to DSS-induced colitis experience reduced colitis severity, including less weight loss, colon damage, improved gut repair, and lowered inflammation. These effects were independent of adaptive immunity and were linked to changes in monocyte function.*
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> *The authors show that the BG-trained monocytes not only help control inflammation but confer non-specific protection against experimental infections (Salmonella), suggesting the involvement of trained immunity (TI) mechanisms. Using single-cell RNA sequencing, they map the transcriptional changes in these cells and show enhanced differentiation of monocytes into reparative CX3CR1$^{+}$ macrophages. Importantly, these protective effects were transferable to other mice via adoptive cell transfer and bone marrow transplantation, suggesting that the innate immune system had been reprogrammed at the level of stem/progenitor cells.*
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> *Overall, this study provides evidence that TI, often associated with heightened inflammatory programs, can also promote tissue repair and resolution of inflammation. Moreover, this BG-induced functional reprogramming can be further harnessed to treat chronic inflammatory disorders like IBD.*
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> *Strengths:*
>
> *(1) The authors use advanced experimental approaches to explore the potential therapeutic use of myeloid reprogramming by β-glucan in IBD.*
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> *(2) The authors follow a data-to-function approach, integrating bulk and single-cell RNA sequencing with in vivo functional validation to support their conclusions.*
>
> *(3) The study adds to the growing evidence that TI is not a singular pro-inflammatory program, but can adopt distinct functional states, including anti-inflammatory and reparative phenotypes, depending on the context.*

We are grateful for the reviewer's positive assessment of our study and recognition of its translational implications. We particularly appreciate the acknowledgment that our work expands the therapeutic potential of β-glucan–mediated trained immunity in ameliorating colitis.

> *Weaknesses:*
>
> *(1) The epigenetic and metabolic basis of TI is not explored, which weakens the mechanistic claim of TI. This is especially relevant given that a novel reparative, anti-*

> *inflammatory TI program is proposed.*

We appreciate the reviewer's valuable comment highlighting the importance of the epigenetic and metabolic basis of TI in providing mechanistic insight. While previous studies, including work from our group (S.-C. Cheng), have extensively characterized the epigenetic and metabolic signatures of monocytes from BG-trained mice—primarily in the context of inflammatory genes—we acknowledge that these aspects are not directly addressed in our current manuscript.

To strengthen the mechanistic component, we plan to: 1. Reanalyze relevant public datasets, focusing on pathways related to reparative and antibacterial function. 2. Perform monocyte ATAC-seq in our current model to validate the epigenetic changes in these pathways.

> *(2) The absence of a BG-only group limits interpretation of the results. Since the authors report tissue-level effects such as enhanced mucosal repair and transcriptional shifts in intestinal macrophages (colonic RNA-Seq), it is important to rule out whether BG alone could influence the gut independently of DSS-induced inflammation.*
>
> *Without a BG-only control, it is hard to distinguish a true trained response from a potential modulation caused directly by BG.*

We thank the reviewer for this important suggestion. Although we did not perform qPCR for mucosal repair genes in Figure S1C and Figure S1D, our colon RNA-seq analysis in Figure 5G included a BG-only control group (Colitis_d0). The results from this group indicate that BG preconditioning alone does not alter baseline expression of colon mucosal repair genes, supporting the conclusion that the observed effects occur in the context of DSS-induced inflammation.

> *(3) Although monocyte transfer experiments show protection in colitis, the fate of the transferred cells is not described (e.g., homing or differentiation into Cx3cr1$^+$ macrophage subsets). This weakens the link between specific monocyte subsets and the observed phenotype.*
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> *(4) While scRNA-seq reveals distinct monocyte/macrophage subclusters (Mono1-3.), their specific functional roles remain speculative. The authors assign reparative or antimicrobial functions based on transcriptional signatures, but do not perform causal experiments (depletion or in vitro assays). The biological roles of these cells remain correlative.*

We agree that the functional role of CX3CR1$^+$ macrophages is not comprehensively validated and is currently inferred from scRNA-seq clustering. While our flow cytometry data show increased CX3CR1$^+$ macrophages in the BG-TI group, and our CCR2 KO and monocyte adoptive transfer experiments indicate these macrophages are monocyte-derived, we lack direct depletion experiments due to the unavailability of effective depletion antibodies for this subset.

We acknowledge this as a limitation and will clarify in the Discussion that our conclusions regarding CX3CR1$^+$ macrophage function are based on transcriptional profiling and association with protective phenotypes, rather than direct causal evidence.

> *(5) While Rag1$^{-/-}$ mice were used to rule out adaptive immunity, the potential role of innate lymphoid cells (ILCs), particularly ILC2s and ILC3s, which are known to promote mucosal repair (PMID: 27484190IF: 7.6 Q1 IF: 7.6 Q1 IF: 7.6 Q1 IF: 7.6 Q1 IF: 7.6 Q1 IF: 7.6 Q1 ), was not explored. Given the reparative phenotype observed, the contribution of ILCs remains a confounding factor.*

We appreciate the reviewer's valuable comment regarding the potential role of ILCs in the observed mucosal repair. Indeed, in examining the BG-trained immunity effect, the contribution of ILCs was not evaluated. We will explicitly acknowledge in the Discussion that *Rag1* $^{-/-}$ mice retain ILCs (including ILC3s) and that BG-induced activation of these cells remains possible.

The literature (PMID: 21502992; PMID: 32187516) supports a role for ILC3-mediated IL-22 production in tissue repair, which could overlap with our observed effects. However, our monocyte adoptive transfer experiments show that monocytes alone can alleviate DSS-induced colitis, suggesting a dominant role for monocytes in this context. Nonetheless, we will make it clear that ILC contributions cannot be excluded.

> ***Reviewer #3 (Public review):***
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> *Summary:*
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> *In the present work, Yinyin Lv et al offer evidence for the therapeutic potential of trained immunity in the context of inflammatory bowel disease (IBD). Prior research has demonstrated that innate cells pre-treated (trained) with β-glucan show an enhanced pro-inflammatory response upon a second challenge.*
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We thank the reviewer for their positive evaluation and constructive feedback on our manuscript.

> *Weaknesses:*
>
> *However, FPKM is not ideal for between-sample comparisons due to its within-sample normalization approach. Best practices recommend using raw counts (with DESeq2) for more robust statistical inference.*

We appreciate the reminder about best practices for RNA-seq analysis. We apologize for the inaccurate description in the Materials and Methods section. For all differential expression analyses, we have in fact used raw count data as input for DESeq2. FPKM values were only used for visualization purposes, such as in heatmaps and clustering analyses. We will correct this description in the revised manuscript to accurately reflect our analysis workflow.

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